Chapter 6

Dynamic programming

In the preceding chapters we have seen some elegant design principles—such as divide-and-conquer, graph exploration, and greedy choice—that yield definitive algorithms for a variety of important computational tasks. The drawback of these tools is that they can only be used on very specific types of problems. We now turn to the two *sledgehammers* of the algorithms craft, *dynamic programming* and *linear programming*, techniques of very broad applicability that can be invoked when more specialized methods fail. Predictably, this generality often comes with a cost in efficiency.

6.1 Shortest paths in dags, revisited

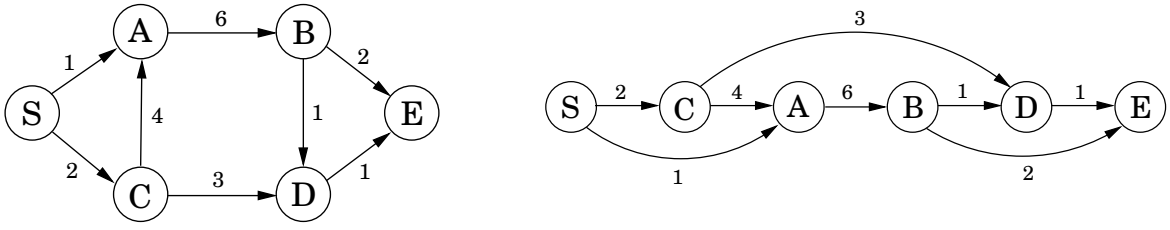
At the conclusion of our study of shortest paths (Chapter 4), we observed that the problem is especially easy in directed acyclic graphs (dags). Let's recapitulate this case, because it lies at the heart of dynamic programming.

The special distinguishing feature of a dag is that its nodes can be *linearized*; that is, they can be arranged on a line so that all edges go from left to right (Figure 6.1). To see why this helps with shortest paths, suppose we want to figure out distances from node S to the other nodes. For concreteness, let's focus on node D. The only way to get to it is through its predecessors, B or C; so to find the shortest path to D, we need only compare these two routes:

$$\texttt{dist}(D) = \min\{\texttt{dist}(B) + 1, \texttt{dist}(C) + 3\}.$$

A similar relation can be written for every node. If we compute these `dist` values in the

Figure 6.1 A dag and its linearization (topological ordering).



left-to-right order of Figure 6.1, we can always be sure that by the time we get to a node v, we already have all the information we need to compute $\texttt{dist}(v)$. We are therefore able to compute all distances in a single pass:

```
initialize all dist(·) values to ∞
dist(s) = 0
for each v ∈ V\{s}, in linearized order:
    dist(v) = min_{(u,v)∈E}{dist(u) + l(u, v)}
```

Notice that this algorithm is solving a collection of *subproblems*, $\{\texttt{dist}(u) : u \in V\}$. We start with the smallest of them, $\texttt{dist}(s)$, since we immediately know its answer to be 0. We then proceed with progressively "larger" subproblems—distances to vertices that are further and further along in the linearization—where we are thinking of a subproblem as large if we need to have solved a lot of other subproblems before we can get to it.

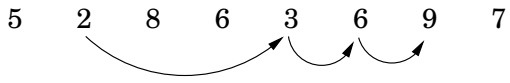
This is a very general technique. At each node, we compute some function of the values of the node's predecessors. It so happens that our particular function is a minimum of sums, but we could just as well make it a *maximum*, in which case we would get *longest* paths in the dag. Or we could use a product instead of a sum inside the brackets, in which case we would end up computing the path with the smallest product of edge lengths.

Dynamic programming is a very powerful algorithmic paradigm in which a problem is solved by identifying a collection of subproblems and tackling them one by one, smallest first, using the answers to small problems to help figure out larger ones, until the whole lot of them is solved. In dynamic programming we are not given a dag; the dag is *implicit*. Its nodes are the subproblems we define, and its edges are the dependencies between the subproblems: if to solve subproblem B we need the answer to subproblem A, then there is a (conceptual) edge from A to B. In this case, A is thought of as a smaller subproblem than B—and it will always be smaller, in an obvious sense.

But it's time we saw an example.

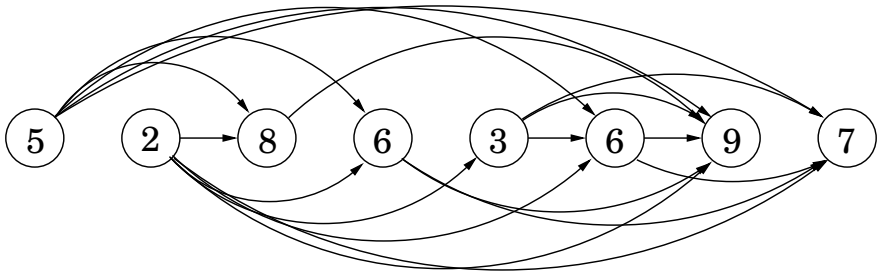
6.2 Longest increasing subsequences

In the *longest increasing subsequence* problem, the input is a sequence of numbers $a_1, \ldots, a_n$. A *subsequence* is any subset of these numbers taken in order, of the form $a_{i_1}, a_{i_2}, \ldots, a_{i_k}$ where $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, and an *increasing* subsequence is one in which the numbers are getting strictly larger. The task is to find the increasing subsequence of greatest length. For instance, the longest increasing subsequence of $5, 2, 8, 6, 3, 6, 9, 7$ is $2, 3, 6, 9$:



In this example, the arrows denote transitions between consecutive elements of the optimal solution. More generally, to better understand the solution space, let's create a graph of *all* permissible transitions: establish a node i for each element a_i, and add directed edges (i, j) whenever it is possible for a_i and a_j to be consecutive elements in an increasing subsequence, that is, whenever $i < j$ and $a_i < a_j$ (Figure 6.2).

Figure 6.2 The dag of increasing subsequences.



Notice that (1) this graph $G = (V, E)$ is a dag, since all edges (i, j) have $i < j$, and (2) there is a one-to-one correspondence between increasing subsequences and paths in this dag. Therefore, our goal is simply to find the longest path in the dag!

Here is the algorithm:

```
for j = 1, 2, ..., n:
    L(j) = 1 + max{L(i) : (i, j) ∈ E}
return max_j L(j)
```

$L(j)$ is the length of the longest path—the longest increasing subsequence—ending at j (plus 1, since strictly speaking we need to count nodes on the path, not edges). By reasoning in the same way as we did for shortest paths, we see that any path to node j must pass through one of its predecessors, and therefore $L(j)$ is 1 plus the maximum $L(\cdot)$ value of these predecessors. If there are no edges into j, we take the maximum over the empty set, zero. And the final answer is the *largest* $L(j)$, since any ending position is allowed.

This is dynamic programming. In order to solve our original problem, we have defined a collection of subproblems $\{L(j) : 1 \leq j \leq n\}$ with the following key property that allows them to be solved in a single pass:

> (*) There is an ordering on the subproblems, and a relation that shows how to solve a subproblem given the answers to "smaller" subproblems, that is, subproblems that appear earlier in the ordering.

In our case, each subproblem is solved using the relation

$$L(j) \;=\; 1 + \max\{L(i) : (i, j) \in E\},$$

an expression which involves only smaller subproblems. How long does this step take? It requires the predecessors of j to be known; for this the adjacency list of the reverse graph G^R, constructible in linear time (recall Exercise 3.5), is handy. The computation of $L(j)$ then takes time proportional to the indegree of j, giving an overall running time linear in $|E|$. This is at most $O(n^2)$, the maximum being when the input array is sorted in increasing order. Thus the dynamic programming solution is both simple and efficient.

There is one last issue to be cleared up: the L-values only tell us the *length* of the optimal subsequence, so how do we recover the subsequence itself? This is easily managed with the

same bookkeeping device we used for shortest paths in Chapter 4. While computing $L(j)$, we should also note down $\texttt{prev}(j)$, the next-to-last node on the longest path to j. The optimal subsequence can then be reconstructed by following these backpointers.

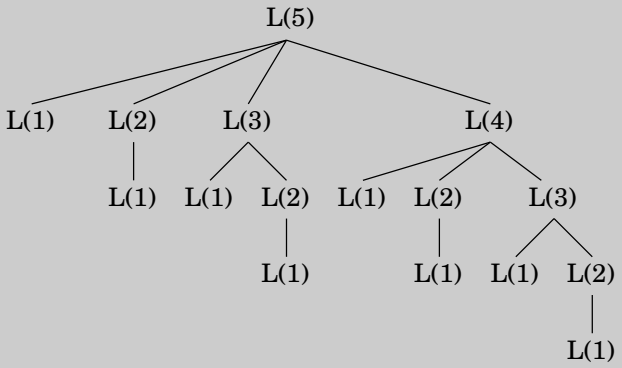
Recursion? No, thanks.

Returning to our discussion of longest increasing subsequences: the formula for $L(j)$ also suggests an alternative, recursive algorithm. Wouldn't that be even simpler?

Actually, recursion is a very bad idea: the resulting procedure would require exponential time! To see why, suppose that the dag contains edges (i, j) for *all* $i < j$—that is, the given sequence of numbers $a_1, a_2, \ldots, a_n$ is sorted. In that case, the formula for subproblem $L(j)$ becomes

$$L(j) = 1 + \max\{L(1), L(2), \ldots, L(j-1)\}.$$

The following figure unravels the recursion for $L(5)$. Notice that the same subproblems get solved over and over again!



For $L(n)$ this tree has exponentially many nodes (can you bound it?), and so a recursive solution is disastrous.

Then why did recursion work so well with divide-and-conquer? The key point is that in divide-and-conquer, a problem is expressed in terms of subproblems that are *substantially smaller*, say half the size. For instance, `mergesort` sorts an array of size n by recursively sorting two subarrays of size $n/2$. Because of this sharp drop in problem size, the full recursion tree has only logarithmic depth and a polynomial number of nodes.

In contrast, in a typical dynamic programming formulation, a problem is reduced to subproblems that are only slightly smaller—for instance, $L(j)$ relies on $L(j-1)$. Thus the full recursion tree generally has polynomial depth and an exponential number of nodes. However, it turns out that most of these nodes are repeats, that there are not too many *distinct* subproblems among them. Efficiency is therefore obtained by explicitly enumerating the distinct subproblems and solving them in the right order.

Programming?

The origin of the term *dynamic programming* has very little to do with writing code. It was first coined by Richard Bellman in the 1950s, a time when computer programming was an esoteric activity practiced by so few people as to not even merit a name. Back then programming meant "planning," and "dynamic programming" was conceived to optimally plan multistage processes. The dag of Figure 6.2 can be thought of as describing the possible ways in which such a process can evolve: each node denotes a state, the leftmost node is the starting point, and the edges leaving a state represent possible actions, leading to different states in the next unit of time.

The etymology of *linear programming*, the subject of Chapter 7, is similar.

6.3 Edit distance

When a spell checker encounters a possible misspelling, it looks in its dictionary for other words that are close by. What is the appropriate notion of closeness in this case?

A natural measure of the distance between two strings is the extent to which they can be *aligned*, or matched up. Technically, an alignment is simply a way of writing the strings one above the other. For instance, here are two possible alignments of SNOWY and SUNNY:

S	–	N	O	W	Y		–	S	N	O	W	–	Y
S	U	N	N	–	Y		S	U	N	–	–	N	Y
Cost: 3							Cost: 5						

The "–" indicates a "gap"; any number of these can be placed in either string. The *cost* of an alignment is the number of columns in which the letters differ. And the *edit distance* between two strings is the cost of their best possible alignment. Do you see that there is no better alignment of SNOWY and SUNNY than the one shown here with a cost of 3?

Edit distance is so named because it can also be thought of as the minimum number of *edits*—insertions, deletions, and substitutions of characters—needed to transform the first string into the second. For instance, the alignment shown on the left corresponds to three edits: insert U, substitute O $\rightarrow$ N, and delete W.

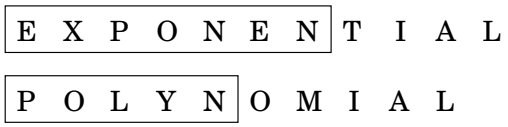
In general, there are so many possible alignments between two strings that it would be terribly inefficient to search through all of them for the best one. Instead we turn to dynamic programming.

A dynamic programming solution

When solving a problem by dynamic programming, the most crucial question is, *What are the subproblems?* As long as they are chosen so as to have the property (*) from page 163. it is an easy matter to write down the algorithm: iteratively solve one subproblem after the other, in order of increasing size.

Our goal is to find the edit distance between two strings $x[1 \cdots m]$ and $y[1 \cdots n]$. What is a good subproblem? Well, it should go part of the way toward solving the whole problem; so how about looking at the edit distance between some *prefix* of the first string, $x[1 \cdots i]$, and some *prefix* of the second, $y[1 \cdots j]$? Call this subproblem $E(i, j)$ (see Figure 6.3). Our final objective, then, is to compute $E(m, n)$.

Figure 6.3 The subproblem $E(7,5)$.



For this to work, we need to somehow express $E(i,j)$ in terms of smaller subproblems. Let's see—what do we know about the best alignment between $x[1\cdots i]$ and $y[1\cdots j]$? Well, its rightmost column can only be one of three things:

$$\begin{matrix} x[i] \\ - \end{matrix} \quad \text{or} \quad \begin{matrix} - \\ y[j] \end{matrix} \quad \text{or} \quad \begin{matrix} x[i] \\ y[j] \end{matrix}$$

The first case incurs a cost of 1 for this particular column, and it remains to align $x[1\cdots i-1]$ with $y[1\cdots j]$. But this is exactly the subproblem $E(i-1,j)$! We seem to be getting somewhere. In the second case, also with cost 1, we still need to align $x[1\cdots i]$ with $y[1\cdots j-1]$. This is again another subproblem, $E(i,j-1)$. And in the final case, which either costs 1 (if $x[i]\neq y[j]$) or 0 (if $x[i]=y[j]$), what's left is the subproblem $E(i-1,j-1)$. In short, we have expressed $E(i,j)$ in terms of three *smaller* subproblems $E(i-1,j)$, $E(i,j-1)$, $E(i-1,j-1)$. We have no idea which of them is the right one, so we need to try them all and pick the best:

$$E(i,j) \;=\; \min\{1+E(i-1,j),\ 1+E(i,j-1),\ \text{diff}(i,j)+E(i-1,j-1)\}$$

where for convenience diff(i,j) is defined to be 0 if $x[i]=y[j]$ and 1 otherwise.

For instance, in computing the edit distance between `EXPONENTIAL` and `POLYNOMIAL`, subproblem $E(4,3)$ corresponds to the prefixes `EXPO` and `POL`. The rightmost column of their best alignment must be one of the following:

$$\begin{matrix} \mathtt{O} \\ - \end{matrix} \quad \mathtt{or} \quad \begin{matrix} - \\ \mathtt{L} \end{matrix} \quad \mathtt{or} \quad \begin{matrix} \mathtt{O} \\ \mathtt{L} \end{matrix}$$

Thus, $E(4,3)=\min\{1+E(3,3),\ 1+E(4,2),\ 1+E(3,2)\}$.

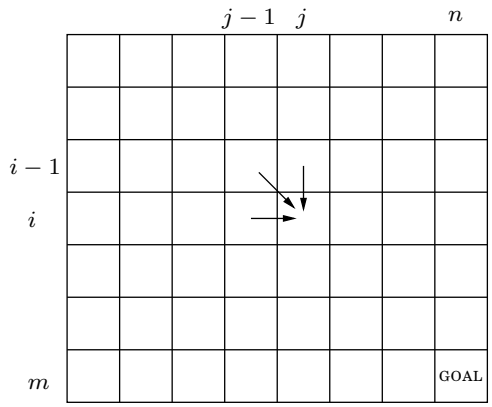
The answers to all the subproblems $E(i,j)$ form a two-dimensional table, as in Figure 6.4. In what order should these subproblems be solved? Any order is fine, as long as $E(i-1,j)$, $E(i,j-1)$, and $E(i-1,j-1)$ are handled *before* $E(i,j)$. For instance, we could fill in the table one row at a time, from top row to bottom row, and moving left to right across each row. Or alternatively, we could fill it in column by column. Both methods would ensure that by the time we get around to computing a particular table entry, all the other entries we need are already filled in.

With both the subproblems and the ordering specified, we are almost done. There just remain the "base cases" of the dynamic programming, the very smallest subproblems. In the present situation, these are $E(0,\cdot)$ and $E(\cdot,0)$, both of which are easily solved. $E(0,j)$ is the edit distance between the 0-length prefix of x, namely the empty string, and the first j letters of y: clearly, j. And similarly, $E(i,0)=i$.

At this point, the algorithm for edit distance basically writes itself.

Figure 6.4 (a) The table of subproblems. Entries $E(i-1, j-1)$, $E(i-1, j)$, and $E(i, j-1)$ are needed to fill in $E(i, j)$. (b) The final table of values found by dynamic programming.

(a)



(b)

		P	O	L	Y	N	O	M	I	A	L
	0	1	2	3	4	5	6	7	8	9	10
E	1	1	2	3	4	5	6	7	8	9	10
X	2	2	2	3	4	5	6	7	8	9	10
P	3	2	3	3	4	5	6	7	8	9	10
O	4	3	2	3	4	5	5	6	7	8	9
N	5	4	3	3	4	4	5	6	7	8	9
E	6	5	4	4	4	5	5	6	7	8	9
N	7	6	5	5	5	4	5	6	7	8	9
T	8	7	6	6	6	5	5	6	7	8	9
I	9	8	7	7	7	6	6	6	6	7	8
A	10	9	8	8	8	7	7	7	7	6	7
L	11	10	9	8	9	8	8	8	8	7	6

```
for i = 0, 1, 2, ..., m:
    E(i, 0) = i
for j = 1, 2, ..., n:
    E(0, j) = j
for i = 1, 2, ..., m:
    for j = 1, 2, ..., n:
        E(i, j) = min{E(i-1, j) + 1, E(i, j-1) + 1, E(i-1, j-1) + diff(i, j)}
return E(m, n)
```

This procedure fills in the table row by row, and left to right within each row. Each entry takes constant time to fill in, so the overall running time is just the size of the table, $O(mn)$.

And in our example, the edit distance turns out to be 6:

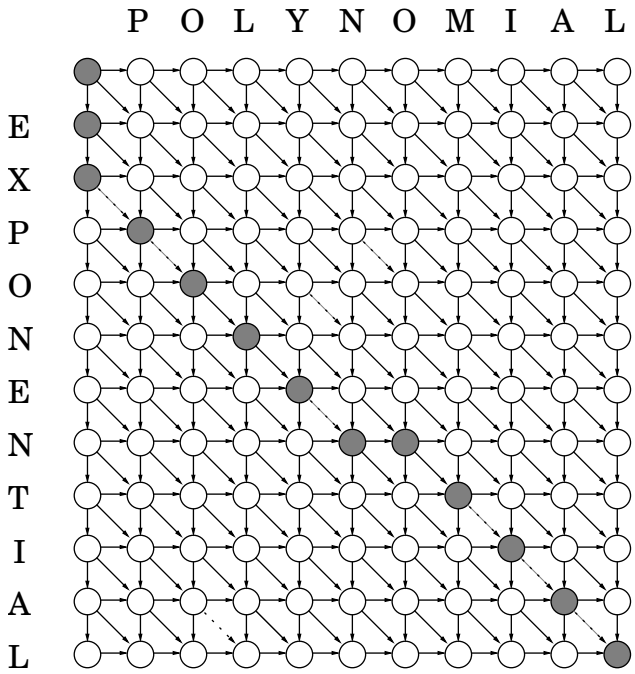
```
E  X  P  O  N  E  N  -  T  I  A  L
-  -  P  O  L  Y  N  O  M  I  A  L
```

The underlying dag

Every dynamic program has an underlying dag structure: think of each node as representing a subproblem, and each edge as a precedence constraint on the order in which the subproblems can be tackled. Having nodes $u_1, \ldots, u_k$ point to v means "subproblem v can only be solved once the answers to $u_1, \ldots, u_k$ are known."

In our present edit distance application, the nodes of the underlying dag correspond to subproblems, or equivalently, to positions (i, j) in the table. Its edges are the precedence constraints, of the form $(i-1, j) \rightarrow (i, j)$, $(i, j-1) \rightarrow (i, j)$, and $(i-1, j-1) \rightarrow (i, j)$ (Figure 6.5). In fact, we can take things a little further and put weights on the edges so that the edit

Figure 6.5 The underlying dag, and a path of length 6.



distances are given by shortest paths in the dag! To see this, set all edge lengths to 1, except for $\{(i-1, j-1) \rightarrow (i, j) : x[i] = y[j]\}$ (shown dotted in the figure), whose length is 0. The final answer is then simply the distance between nodes $s = (0, 0)$ and $t = (m, n)$. One possible shortest path is shown, the one that yields the alignment we found earlier. On this path, each move down is a deletion, each move right is an insertion, and each diagonal move is either a match or a substitution.

By altering the weights on this dag, we can allow generalized forms of edit distance, in which insertions, deletions, and substitutions have different associated costs.

Common subproblems

Finding the right subproblem takes creativity and experimentation. But there are a few standard choices that seem to arise repeatedly in dynamic programming.

i. The input is $x_1, x_2, \ldots, x_n$ and a subproblem is $x_1, x_2, \ldots, x_i$.

$$\boxed{x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6} \quad x_7 \quad x_8 \quad x_9 \quad x_{10}$$

The number of subproblems is therefore linear.

ii. The input is $x_1, \ldots, x_n$, and $y_1, \ldots, y_m$. A subproblem is $x_1, \ldots, x_i$ and $y_1, \ldots, y_j$.

$$\boxed{x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6} \quad x_7 \quad x_8 \quad x_9 \quad x_{10}$$

$$\boxed{y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5} \quad y_6 \quad y_7 \quad y_8$$

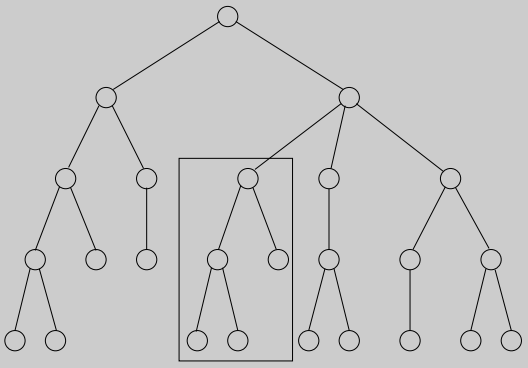
The number of subproblems is $O(mn)$.

iii. The input is $x_1, \ldots, x_n$ and a subproblem is $x_i, x_{i+1}, \ldots, x_j$.

$$x_1 \quad x_2 \quad \boxed{x_3 \quad x_4 \quad x_5 \quad x_6} \quad x_7 \quad x_8 \quad x_9 \quad x_{10}$$

The number of subproblems is $O(n^2)$.

iv. The input is a rooted tree. A subproblem is a rooted subtree.



If the tree has n nodes, how many subproblems are there?

We've already encountered the first two cases, and the others are coming up shortly.

Of mice and men

Our bodies are extraordinary machines: flexible in function, adaptive to new environments, and able to interact and reproduce. All these capabilities are specified by a program unique to each of us, a string that is 3 billion characters long over the alphabet $\{A, C, G, T\}$—our DNA.

The DNA sequences of any two people differ by only about 0.1%. However, this still leaves 3 million positions on which they vary, more than enough to explain the vast range of human diversity. These differences are of great scientific and medical interest—for instance, they might help predict which people are prone to certain diseases.

DNA is a vast and seemingly inscrutable program, but it can be broken down into smaller units that are more specific in their role, rather like subroutines. These are called *genes*. Computers have become a crucial tool in understanding the genes of humans and other organisms, to the extent that *computational genomics* is now a field in its own right. Here are examples of typical questions that arise.

1. When a new gene is discovered, one way to gain insight into its function is to find known genes that match it closely. This is particularly helpful in transferring knowledge from well-studied species, such as mice, to human beings.

   A basic primitive in this search problem is to define an efficiently computable notion of when two strings approximately match. The biology suggests a generalization of edit distance, and dynamic programming can be used to compute it.

   Then there's the problem of searching through the vast thicket of known genes: the database GenBank already has a total length of over 10^{10}, and this number is growing rapidly. The current method of choice is BLAST, a clever combination of algorithmic tricks and biological intuitions that has made it the most widely used software in computational biology.

2. Methods for *sequencing* DNA (that is, determining the string of characters that constitute it) typically only find fragments of 500–700 characters. Billions of these randomly scattered fragments can be generated, but how can they be assembled into a coherent DNA sequence? For one thing, the position of any one fragment in the final sequence is unknown and must be inferred by piecing together overlapping fragments.

   A showpiece of these efforts is the draft of human DNA completed in 2001 by two groups simultaneously: the publicly funded Human Genome Consortium and the private Celera Genomics.

3. When a particular gene has been sequenced in each of several species, can this information be used to reconstruct the evolutionary history of these species?

We will explore these problems in the exercises at the end of this chapter. Dynamic programming has turned out to be an invaluable tool for some of them and for computational biology in general.

6.4 Knapsack

During a robbery, a burglar finds much more loot than he had expected and has to decide what to take. His bag (or "knapsack") will hold a total weight of at most W pounds. There are n items to pick from, of weight $w_1, \ldots, w_n$ and dollar value $v_1, \ldots, v_n$. What's the most valuable combination of items he can fit into his bag?[1]

For instance, take $W = 10$ and

Item	Weight	Value
1	6	\$30
2	3	\$14
3	4	\$16
4	2	\$9

There are two versions of this problem. If there are unlimited quantities of each item available, the optimal choice is to pick item 1 and two of item 4 (total: \$48). On the other hand, if there is one of each item (the burglar has broken into an art gallery, say), then the optimal knapsack contains items 1 and 3 (total: \$46).

As we shall see in Chapter 8, neither version of this problem is likely to have a polynomial-time algorithm. However, using dynamic programming they can both be solved in $O(nW)$ time, which is reasonable when W is small, but is not polynomial since the input size is proportional to $\log W$ rather than W.

Knapsack with repetition

Let's start with the version that allows repetition. As always, the main question in dynamic programming is, what are the subproblems? In this case we can shrink the original problem in two ways: we can either look at smaller knapsack capacities $w \leq W$, or we can look at fewer items (for instance, items $1, 2, \ldots, j$, for $j \leq n$). It usually takes a little experimentation to figure out exactly what works.

The first restriction calls for smaller capacities. Accordingly, define

$$K(w) = \text{maximum value achievable with a knapsack of capacity } w.$$

Can we express this in terms of smaller subproblems? Well, if the optimal solution to $K(w)$ includes item i, then removing this item from the knapsack leaves an optimal solution to $K(w - w_i)$. In other words, $K(w)$ is simply $K(w - w_i) + v_i$, for some i. We don't know which i, so we need to try all possibilities.

$$K(w) = \max_{i: w_i \leq w} \{K(w - w_i) + v_i\},$$

where as usual our convention is that the maximum over an empty set is 0. We're done! The algorithm now writes itself, and it is characteristically simple and elegant.

[1] If this application seems frivolous, replace "weight" with "CPU time" and "only W pounds can be taken" with "only W units of CPU time are available." Or use "bandwidth" in place of "CPU time," etc. The knapsack problem generalizes a wide variety of resource-constrained selection tasks.

```
K(0) = 0
for w = 1 to W:
    K(w) = max{K(w − w_i) + v_i : w_i ≤ w}
return K(W)
```

This algorithm fills in a one-dimensional table of length $W + 1$, in left-to-right order. Each entry can take up to $O(n)$ time to compute, so the overall running time is $O(nW)$.

As always, there is an underlying dag. Try constructing it, and you will be rewarded with a startling insight: this particular variant of knapsack boils down to finding the longest path in a dag!

Knapsack without repetition

On to the second variant: what if repetitions are not allowed? Our earlier subproblems now become completely useless. For instance, knowing that the value $K(w - w_n)$ is very high doesn't help us, because we don't know whether or not item n already got used up in this partial solution. We must therefore refine our concept of a subproblem to carry additional information about the items being used. We add a second parameter, $0 \leq j \leq n$:

$$K(w, j) = \text{maximum value achievable using a knapsack of capacity } w \text{ and items } 1, \ldots, j.$$

The answer we seek is $K(W, n)$.

How can we express a subproblem $K(w, j)$ in terms of smaller subproblems? Quite simple: either item j is needed to achieve the optimal value, or it isn't needed:

$$K(w, j) = \max\{K(w - w_j, j - 1) + v_j, K(w, j - 1)\}.$$

(The first case is invoked only if $w_j \leq w$.) In other words, we can express $K(w, j)$ in terms of subproblems $K(\cdot, j - 1)$.

The algorithm then consists of filling out a two-dimensional table, with $W + 1$ rows and $n + 1$ columns. Each table entry takes just constant time, so even though the table is much larger than in the previous case, the running time remains the same, $O(nW)$. Here's the code.

```
Initialize all K(0, j) = 0 and all K(w, 0) = 0
for j = 1 to n:
   for w = 1 to W:
      if w_j > w:   K(w, j) = K(w, j − 1)
      else:   K(w, j) = max{K(w, j − 1), K(w − w_j, j − 1) + v_j}
return K(W, n)
```

Memoization

In dynamic programming, we write out a recursive formula that expresses large problems in terms of smaller ones and then use it to fill out a table of solution values in a bottom-up manner, from smallest subproblem to largest.

The formula also suggests a recursive algorithm, but we saw earlier that naive recursion can be terribly inefficient, because it solves the same subproblems over and over again. What about a more intelligent recursive implementation, one that remembers its previous invocations and thereby avoids repeating them?

On the knapsack problem (with repetitions), such an algorithm would use a hash table (recall Section 1.5) to store the values of $K(\cdot)$ that had already been computed. At each recursive call requesting some $K(w)$, the algorithm would first check if the answer was already in the table and then would proceed to its calculation only if it wasn't. This trick is called *memoization*:

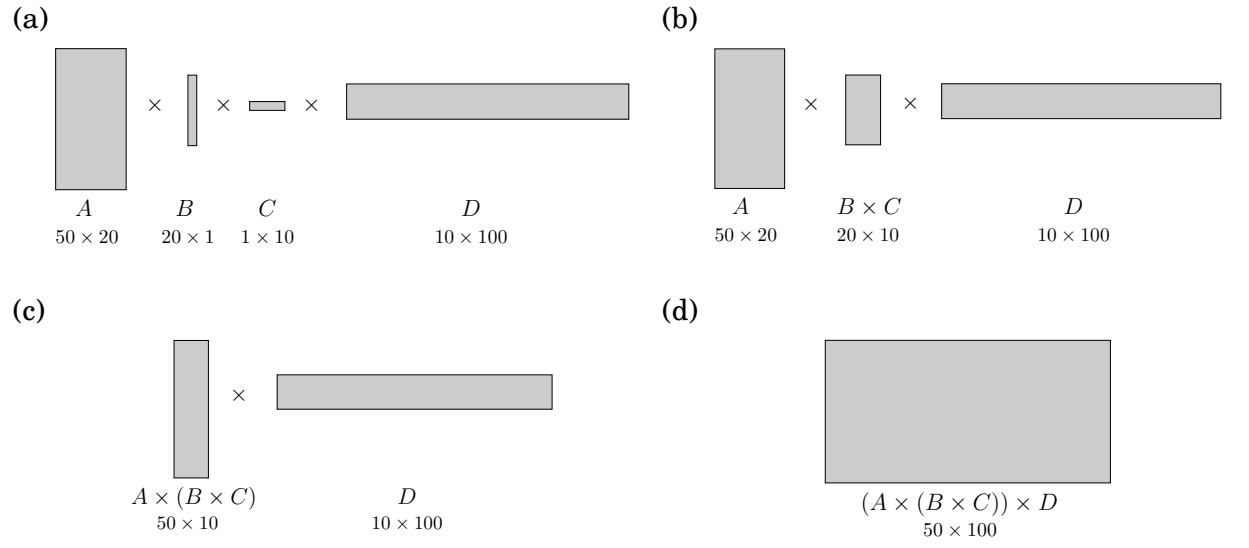
```
A hash table, initially empty, holds values of K(w) indexed by w

function knapsack(w)
if w is in hash table:  return K(w)
K(w) = max{knapsack(w − w_i) + v_i : w_i ≤ w}
insert K(w) into hash table, with key w
return K(w)
```

Since this algorithm never repeats a subproblem, its running time is $O(nW)$, just like the dynamic program. However, the constant factor in this big-O notation is substantially larger because of the overhead of recursion.

In some cases, though, memoization pays off. Here's why: dynamic programming automatically solves every subproblem that could conceivably be needed, while memoization only ends up solving the ones that are actually used. For instance, suppose that W and all the weights w_i are multiples of 100. Then a subproblem $K(w)$ is useless if 100 does not divide w. The memoized recursive algorithm will never look at these extraneous table entries.

Figure 6.6 $A \times B \times C \times D = (A \times (B \times C)) \times D$.



6.5 Chain matrix multiplication

Suppose that we want to multiply four matrices, $A \times B \times C \times D$, of dimensions 50×20, 20×1, 1×10, and 10×100, respectively (Figure 6.6). This will involve iteratively multiplying two matrices at a time. Matrix multiplication is not *commutative* (in general, $A \times B \neq B \times A$), but it is *associative*, which means for instance that $A \times (B \times C) = (A \times B) \times C$. Thus we can compute our product of four matrices in many different ways, depending on how we parenthesize it. Are some of these better than others?

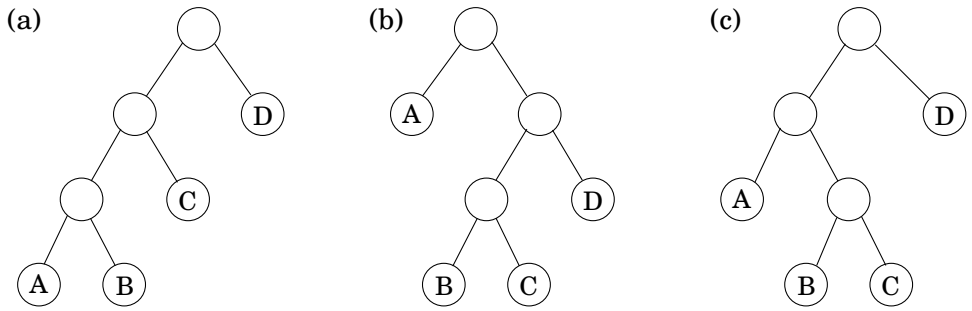
Multiplying an $m \times n$ matrix by an $n \times p$ matrix takes mnp multiplications, to a good enough approximation. Using this formula, let's compare several different ways of evaluating $A \times B \times C \times D$:

Parenthesization	Cost computation	Cost
$A \times ((B \times C) \times D)$	$20 \cdot 1 \cdot 10 + 20 \cdot 10 \cdot 100 + 50 \cdot 20 \cdot 100$	$120,200$
$(A \times (B \times C)) \times D$	$20 \cdot 1 \cdot 10 + 50 \cdot 20 \cdot 10 + 50 \cdot 10 \cdot 100$	$60,200$
$(A \times B) \times (C \times D)$	$50 \cdot 20 \cdot 1 + 1 \cdot 10 \cdot 100 + 50 \cdot 1 \cdot 100$	$7,000$

As you can see, the order of multiplications makes a big difference in the final running time! Moreover, the natural *greedy* approach, to always perform the cheapest matrix multiplication available, leads to the second parenthesization shown here and is therefore a failure.

How do we determine the optimal order, if we want to compute $A_1 \times A_2 \times \cdots \times A_n$, where the A_i's are matrices with dimensions $m_0 \times m_1, m_1 \times m_2, \ldots, m_{n-1} \times m_n$, respectively? The first thing to notice is that a particular parenthesization can be represented very naturally by a binary tree in which the individual matrices correspond to the leaves, the root is the final product, and interior nodes are intermediate products (Figure 6.7). The possible orders in which to do the multiplication correspond to the various full binary trees with n leaves, whose

Figure 6.7 (a) $((A \times B) \times C) \times D$; (b) $A \times ((B \times C) \times D)$; (c) $(A \times (B \times C)) \times D$.



number is exponential in n (Exercise 2.13). We certainly cannot try each tree, and with brute force thus ruled out, we turn to dynamic programming.

The binary trees of Figure 6.7 are suggestive: for a tree to be optimal, its subtrees must also be optimal. What are the subproblems corresponding to the subtrees? They are products of the form $A_i \times A_{i+1} \times \cdots \times A_j$. Let's see if this works: for $1 \leq i \leq j \leq n$, define

$$C(i,j) = \text{minimum cost of multiplying } A_i \times A_{i+1} \times \cdots \times A_j.$$

The size of this subproblem is the number of matrix multiplications, $|j - i|$. The smallest subproblem is when $i = j$, in which case there's nothing to multiply, so $C(i,i) = 0$. For $j > i$, consider the optimal subtree for $C(i,j)$. The first branch in this subtree, the one at the top, will split the product in two pieces, of the form $A_i \times \cdots \times A_k$ and $A_{k+1} \times \cdots \times A_j$, for some k between i and j. The cost of the subtree is then the cost of these two partial products, plus the cost of combining them: $C(i,k) + C(k+1,j) + m_{i-1} \cdot m_k \cdot m_j$. And we just need to find the splitting point k for which this is smallest:

$$C(i,j) = \min_{i \leq k < j} \{C(i,k) + C(k+1,j) + m_{i-1} \cdot m_k \cdot m_j\}.$$

We are ready to code! In the following, the variable s denotes subproblem size.

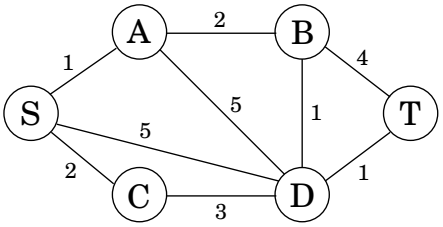
```
for i = 1 to n:   C(i,i) = 0
for s = 1 to n-1:
  for i = 1 to n-s:
    j = i + s
    C(i,j) = min{C(i,k) + C(k+1,j) + m_{i-1} · m_k · m_j : i ≤ k < j}
return C(1,n)
```

The subproblems constitute a two-dimensional table, each of whose entries takes $O(n)$ time to compute. The overall running time is thus $O(n^3)$.

6.6 Shortest paths

We started this chapter with a dynamic programming algorithm for the elementary task of finding the shortest path in a dag. We now turn to more sophisticated shortest-path problems

Figure 6.8 We want a path from s to t that is both short *and* has few edges.



and see how these too can be accommodated by our powerful algorithmic technique.

Shortest reliable paths

Life is complicated, and abstractions such as graphs, edge lengths, and shortest paths rarely capture the whole truth. In a communications network, for example, even if edge lengths faithfully reflect transmission delays, there may be other considerations involved in choosing a path. For instance, each extra edge in the path might be an extra "hop" fraught with uncertainties and dangers of packet loss. In such cases, we would like to avoid paths with too many edges. Figure 6.8 illustrates this problem with a graph in which the shortest path from S to T has four edges, while there is another path that is a little longer but uses only two edges. If four edges translate to prohibitive unreliability, we may have to choose the latter path.

Suppose then that we are given a graph G with lengths on the edges, along with two nodes s and t and an integer k, and we want the shortest path from s to t *that uses at most k edges*.

Is there a quick way to adapt Dijkstra's algorithm to this new task? Not quite: that algorithm focuses on the length of each shortest path without "remembering" the number of hops in the path, which is now a crucial piece of information.

In dynamic programming, the trick is to choose subproblems so that all vital information is remembered and carried forward. In this case, let us define, for each vertex v and each integer $i \leq k$, $\texttt{dist}(v, i)$ to be *the length of the shortest path from s to v that uses i edges*. The starting values $\texttt{dist}(v, 0)$ are ∞ for all vertices except s, for which it is 0. And the general update equation is, naturally enough,

$$\texttt{dist}(v, i) \;=\; \min_{(u,v)\in E} \{\texttt{dist}(u, i-1) + \ell(u, v)\}.$$

Need we say more?

All-pairs shortest paths

What if we want to find the shortest path not just between s and t but between *all* pairs of vertices? One approach would be to execute our general shortest-path algorithm from Section 4.6.1 (since there may be negative edges) $|V|$ times, once for each starting node. The total running time would then be $O(|V|^2|E|)$. We'll now see a better alternative, the $O(|V|^3)$ dynamic programming-based *Floyd-Warshall algorithm*.

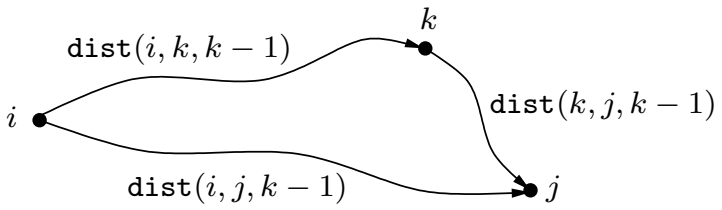
Is there is a good subproblem for computing distances between all pairs of vertices in a graph? Simply solving the problem for more and more pairs or starting points is unhelpful,

because it leads right back to the $O(|V|^2|E|)$ algorithm.

One idea comes to mind: the shortest path $u \to w_1 \to \cdots \to w_l \to v$ between u and v uses some number of intermediate nodes—possibly none. Suppose we disallow intermediate nodes altogether. Then we can solve all-pairs shortest paths at once: the shortest path from u to v is simply the direct edge (u, v), if it exists. What if we now gradually expand the *set of permissible intermediate nodes*? We can do this one node at a time, updating the shortest path lengths at each stage. Eventually this set grows to all of V, at which point all vertices are allowed to be on all paths, and we have found the true shortest paths between vertices of the graph!

More concretely, number the vertices in V as $\{1, 2, \ldots, n\}$, and let $\texttt{dist}(i, j, k)$ denote the length of the shortest path from i to j in which only nodes $\{1, 2, \ldots, k\}$ can be used as intermediates. Initially, $\texttt{dist}(i, j, 0)$ is the length of the direct edge between i and j, if it exists, and is ∞ otherwise.

What happens when we expand the intermediate set to include an extra node k? We must reexamine all pairs i, j and check whether using k as an intermediate point gives us a shorter path from i to j. But this is easy: a shortest path from i to j that uses k along with possibly other lower-numbered intermediate nodes goes through k just once (why? because we assume that there are no negative cycles). And we have already calculated the length of the shortest path from i to k and from k to j using only lower-numbered vertices:



Thus, using k gives us a shorter path from i to j if and only if

$$\texttt{dist}(i, k, k-1) + \texttt{dist}(k, j, k-1) \;<\; \texttt{dist}(i, j, k-1),$$

in which case $\texttt{dist}(i, j, k)$ should be updated accordingly.

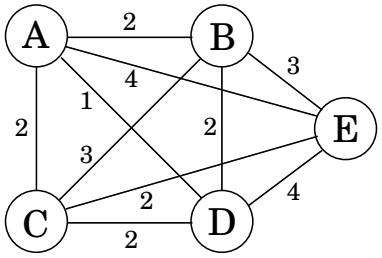
Here is the Floyd-Warshall algorithm—and as you can see, it takes $O(|V|^3)$ time.

```
for i = 1 to n:
    for j = 1 to n:
        dist(i, j, 0) = ∞
for all (i, j) ∈ E:
    dist(i, j, 0) = ℓ(i, j)
for k = 1 to n:
    for i = 1 to n:
        for j = 1 to n:
            dist(i, j, k) = min{dist(i, k, k − 1) + dist(k, j, k − 1),  dist(i, j, k − 1)}
```

The traveling salesman problem

A traveling salesman is getting ready for a big sales tour. Starting at his hometown, suitcase in hand, he will conduct a journey in which each of his target cities is visited exactly once

Figure 6.9 The optimal traveling salesman tour has length 10.



before he returns home. Given the pairwise distances between cities, what is the best order in which to visit them, so as to minimize the overall distance traveled?

Denote the cities by $1, \ldots, n$, the salesman's hometown being 1, and let $D = (d_{ij})$ be the matrix of intercity distances. The goal is to design a tour that starts and ends at 1, includes all other cities exactly once, and has minimum total length. Figure 6.9 shows an example involving five cities. Can you spot the optimal tour? Even in this tiny example, it is tricky for a human to find the solution; imagine what happens when hundreds of cities are involved.

It turns out this problem is also difficult for computers. In fact, the traveling salesman problem (TSP) is one of the most notorious computational tasks. There is a long history of attempts at solving it, a long saga of failures and partial successes, and along the way, major advances in algorithms and complexity theory. The most basic piece of bad news about the TSP, which we will better understand in Chapter 8, is that it is highly unlikely to be solvable in polynomial time.

How long does it take, then? Well, the brute-force approach is to evaluate every possible tour and return the best one. Since there are $(n-1)!$ possibilities, this strategy takes $O(n!)$ time. We will now see that dynamic programming yields a much faster solution, though not a polynomial one.

What is the appropriate subproblem for the TSP? Subproblems refer to partial solutions, and in this case the most obvious partial solution is the initial portion of a tour. Suppose we have started at city 1 as required, have visited a few cities, and are now in city j. What information do we need in order to extend this partial tour? We certainly need to know j, since this will determine which cities are most convenient to visit next. And we also need to know all the cities visited so far, so that we don't repeat any of them. Here, then, is an appropriate subproblem.

> For a subset of cities $S \subseteq \{1, 2, \ldots, n\}$ that includes 1, and $j \in S$, let $C(S, j)$ be the length of the shortest path visiting each node in S exactly once, starting at 1 and ending at j.

When $|S| > 1$, we define $C(S, 1) = \infty$ since the path cannot both start and end at 1.

Now, let's express $C(S, j)$ in terms of smaller subproblems. We need to start at 1 and end at j; what should we pick as the second-to-last city? It has to be some $i \in S$, so the overall path length is the distance from 1 to i, namely, $C(S - \{j\}, i)$, plus the length of the final edge,

d_{ij}. We must pick the best such i:

$$C(S, j) = \min_{i \in S: i \neq j} C(S - \{j\}, i) + d_{ij}.$$

The subproblems are ordered by $|S|$. Here's the code.

```
C({1}, 1) = 0
for s = 2 to n:
    for all subsets S ⊆ {1, 2, ..., n} of size s and containing 1:
        C(S, 1) = ∞
        for all j ∈ S, j ≠ 1:
            C(S, j) = min{C(S − {j}, i) + d_ij : i ∈ S, i ≠ j}
return min_j C({1, ..., n}, j) + d_j1
```

There are at most $2^n \cdot n$ subproblems, and each one takes linear time to solve. The total running time is therefore $O(n^2 2^n)$.

On time and memory

The amount of time it takes to run a dynamic programming algorithm is easy to discern from the dag of subproblems: in many cases *it is just the total number of edges in the dag!* All we are really doing is visiting the nodes in linearized order, examining each node's inedges, and, most often, doing a constant amount of work per edge. By the end, each edge of the dag has been examined once.

But how much computer *memory* is required? There is no simple parameter of the dag characterizing this. It is certainly possible to do the job with an amount of memory proportional to the number of vertices (subproblems), but we can usually get away with much less. The reason is that the value of a particular subproblem only needs to be remembered until the larger subproblems depending on it have been solved. Thereafter, the memory it takes up can be released for reuse.

For example, in the Floyd-Warshall algorithm the value of $\texttt{dist}(i, j, k)$ is not needed once the $\texttt{dist}(\cdot, \cdot, k+1)$ values have been computed. Therefore, we only need two $|V| \times |V|$ arrays to store the `dist` values, one for odd values of k and one for even values: when computing $\texttt{dist}(i, j, k)$, we overwrite $\texttt{dist}(i, j, k-2)$.

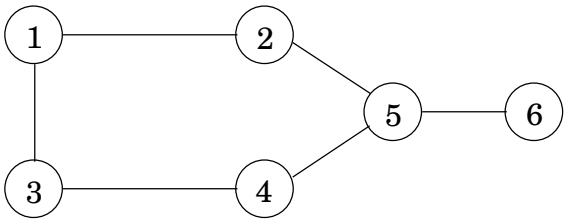
(And let us not forget that, as always in dynamic programming, we also need one more array, $\texttt{prev}(i, j)$, storing the next to last vertex in the current shortest path from i to j, a value that must be updated with $\texttt{dist}(i, j, k)$. We omit this mundane but crucial bookkeeping step from our dynamic programming algorithms.)

Can you see why the edit distance dag in Figure 6.5 only needs memory proportional to the length of the shorter string?

6.7 Independent sets in trees

A subset of nodes $S \subset V$ is an *independent set* of graph $G = (V, E)$ if there are no edges between them. For instance, in Figure 6.10 the nodes $\{1, 5\}$ form an independent set, but

Figure 6.10 The largest independent set in this graph has size 3.



nodes $\{1, 4, 5\}$ do not, because of the edge between 4 and 5. The largest independent set is $\{2, 3, 6\}$.

Like several other problems we have seen in this chapter (knapsack, traveling salesman), finding the largest independent set in a graph is believed to be intractable. However, when the graph happens to be a *tree*, the problem can be solved in linear time, using dynamic programming. And what are the appropriate subproblems? Already in the chain matrix multiplication problem we noticed that the layered structure of a tree provides a natural definition of a subproblem—as long as one node of the tree has been identified as a root.

So here's the algorithm: Start by rooting the tree at any node r. Now, each node defines a subtree—the one hanging from it. This immediately suggests subproblems:

$$I(u) \;=\; \text{size of largest independent set of subtree hanging from } u.$$

Our final goal is $I(r)$.

Dynamic programming proceeds as always from smaller subproblems to larger ones, that is to say, bottom-up in the rooted tree. Suppose we know the largest independent sets for all subtrees below a certain node u; in other words, suppose we know $I(w)$ for all descendants w of u. How can we compute $I(u)$? Let's split the computation into two cases: any independent set either includes u or it doesn't (Figure 6.11).

$$I(u) \;=\; \max\left\{1 + \sum_{\text{grandchildren } w \text{ of } u} I(w), \sum_{\text{children } w \text{ of } u} I(w)\right\}.$$

If the independent set includes u, then we get one point for it, but we aren't allowed to include the children of u—therefore we move on to the grandchildren. This is the first case in the formula. On the other hand, if we don't include u, then we don't get a point for it, but we can move on to its children.

The number of subproblems is exactly the number of vertices. With a little care, the running time can be made linear, $O(|V| + |E|)$.

Chapter 6

Dynamic Programming

We began our study of algorithmic techniques with greedy algorithms, which in some sense form the most natural approach to algorithm design. Faced with a new computational problem, we've seen that it's not hard to propose multiple possible greedy algorithms; the challenge is then to determine whether any of these algorithms provides a correct solution to the problem in all cases.

The problems we saw in Chapter 4 were all unified by the fact that, in the end, there really was a greedy algorithm that worked. Unfortunately, this is far from being true in general; for most of the problems that one encounters, the real difficulty is not in determining which of several greedy strategies is the right one, but in the fact that there is *no* natural greedy algorithm that works. For such problems, it is important to have other approaches at hand. Divide and conquer can sometimes serve as an alternative approach, but the versions of divide and conquer that we saw in the previous chapter are often not strong enough to reduce exponential brute-force search down to polynomial time. Rather, as we noted in Chapter 5, the applications there tended to reduce a running time that was unnecessarily large, but already polynomial, down to a faster running time.

We now turn to a more powerful and subtle design technique, *dynamic programming*. It will be easier to say exactly what characterizes dynamic programming after we've seen it in action, but the basic idea is drawn from the intuition behind divide and conquer and is essentially the opposite of the greedy strategy: one implicitly explores the space of all possible solutions, by carefully decomposing things into a series of *subproblems*, and then building up correct solutions to larger and larger subproblems. In a way, we can thus view dynamic programming as operating dangerously close to the edge of

brute-force search: although it's systematically working through the exponentially large set of possible solutions to the problem, it does this without ever examining them all explicitly. It is because of this careful balancing act that dynamic programming can be a tricky technique to get used to; it typically takes a reasonable amount of practice before one is fully comfortable with it.

With this in mind, we now turn to a first example of dynamic programming: the Weighted Interval Scheduling Problem that we defined back in Section 1.2. We are going to develop a dynamic programming algorithm for this problem in two stages: first as a recursive procedure that closely resembles brute-force search; and then, by reinterpreting this procedure, as an iterative algorithm that works by building up solutions to larger and larger subproblems.

6.1 Weighted Interval Scheduling: A Recursive Procedure

We have seen that a particular greedy algorithm produces an optimal solution to the Interval Scheduling Problem, where the goal is to accept as large a set of nonoverlapping intervals as possible. The Weighted Interval Scheduling Problem is a strictly more general version, in which each interval has a certain *value* (or *weight*), and we want to accept a set of maximum value.

Designing a Recursive Algorithm

Since the original Interval Scheduling Problem is simply the special case in which all values are equal to 1, we know already that most greedy algorithms will not solve this problem optimally. But even the algorithm that worked before (repeatedly choosing the interval that ends earliest) is no longer optimal in this more general setting, as the simple example in Figure 6.1 shows.

Indeed, no natural greedy algorithm is known for this problem, which is what motivates our switch to dynamic programming. As discussed above, we will begin our introduction to dynamic programming with a recursive type of algorithm for this problem, and then in the next section we'll move to a more iterative method that is closer to the style we use in the rest of this chapter.

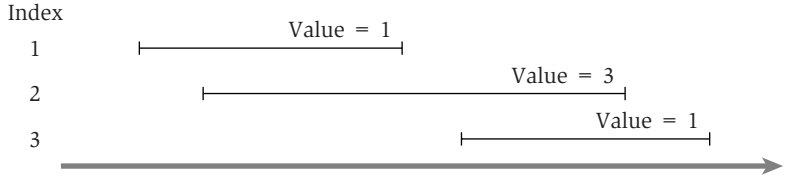


Figure 6.1 A simple instance of weighted interval scheduling.

We use the notation from our discussion of Interval Scheduling in Section 1.2. We have n requests labeled $1, \ldots, n$, with each request i specifying a start time s_i and a finish time f_i. Each interval i now also has a *value*, or *weight* v_i. Two intervals are *compatible* if they do not overlap. The goal of our current problem is to select a subset $S \subseteq \{1, \ldots, n\}$ of mutually compatible intervals, so as to maximize the sum of the values of the selected intervals, $\sum_{i \in S} v_i$.

Let's suppose that the requests are sorted in order of nondecreasing finish time: $f_1 \leq f_2 \leq \cdots \leq f_n$. We'll say a request i comes *before* a request j if $i < j$. This will be the natural left-to-right order in which we'll consider intervals. To help in talking about this order, we define $p(j)$, for an interval j, to be the largest index $i < j$ such that intervals i and j are disjoint. In other words, i is the leftmost interval that ends before j begins. We define $p(j) = 0$ if no request $i < j$ is disjoint from j. An example of the definition of $p(j)$ is shown in Figure 6.2.

Now, given an instance of the Weighted Interval Scheduling Problem, let's consider an optimal solution $\mathcal{O}$, ignoring for now that we have no idea what it is. Here's something completely obvious that we can say about $\mathcal{O}$: either interval n (the last one) belongs to $\mathcal{O}$, or it doesn't. Suppose we explore both sides of this dichotomy a little further. If $n \in \mathcal{O}$, then clearly no interval indexed strictly between $p(n)$ and n can belong to $\mathcal{O}$, because by the definition of $p(n)$, we know that intervals $p(n) + 1, p(n) + 2, \ldots, n - 1$ all overlap interval n. Moreover, if $n \in \mathcal{O}$, then $\mathcal{O}$ must include an *optimal* solution to the problem consisting of requests $\{1, \ldots, p(n)\}$—for if it didn't, we could replace $\mathcal{O}$'s choice of requests from $\{1, \ldots, p(n)\}$ with a better one, with no danger of overlapping request n.

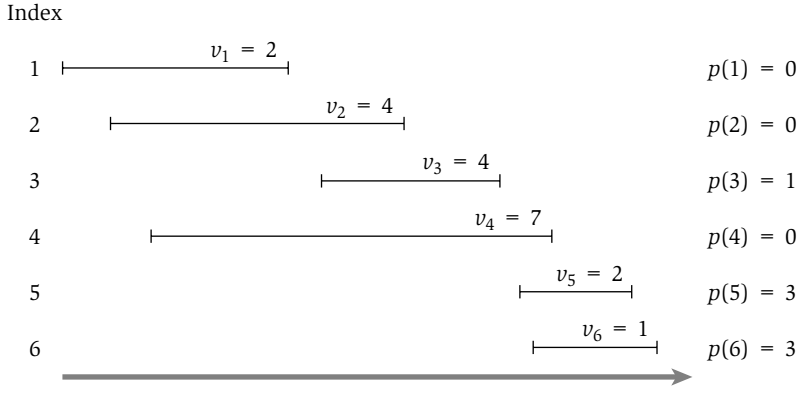


Figure 6.2 An instance of weighted interval scheduling with the functions $p(j)$ defined for each interval j.

On the other hand, if $n \notin \mathcal{O}$, then $\mathcal{O}$ is simply equal to the optimal solution to the problem consisting of requests $\{1, \ldots, n-1\}$. This is by completely analogous reasoning: we're assuming that $\mathcal{O}$ does not include request n; so if it does not choose the optimal set of requests from $\{1, \ldots, n-1\}$, we could replace it with a better one.

All this suggests that finding the optimal solution on intervals $\{1, 2, \ldots, n\}$ involves looking at the optimal solutions of smaller problems of the form $\{1, 2, \ldots, j\}$. Thus, for any value of j between 1 and n, let $\mathcal{O}_j$ denote the optimal solution to the problem consisting of requests $\{1, \ldots, j\}$, and let $\text{OPT}(j)$ denote the value of this solution. (We define $\text{OPT}(0) = 0$, based on the convention that this is the optimum over an empty set of intervals.) The optimal solution we're seeking is precisely $\mathcal{O}_n$, with value $\text{OPT}(n)$. For the optimal solution $\mathcal{O}_j$ on $\{1, 2, \ldots, j\}$, our reasoning above (generalizing from the case in which $j = n$) says that either $j \in \mathcal{O}_j$, in which case $\text{OPT}(j) = v_j + \text{OPT}(p(j))$, or $j \notin \mathcal{O}_j$, in which case $\text{OPT}(j) = \text{OPT}(j-1)$. Since these are precisely the two possible choices ($j \in \mathcal{O}_j$ or $j \notin \mathcal{O}_j$), we can further say that

(6.1) $$\text{OPT}(j) = \max(v_j + \text{OPT}(p(j)), \text{OPT}(j-1)).$$

And how do we decide whether n belongs to the optimal solution $\mathcal{O}_j$? This too is easy: it belongs to the optimal solution if and only if the first of the options above is at least as good as the second; in other words,

(6.2) *Request j belongs to an optimal solution on the set $\{1, 2, \ldots, j\}$ if and only if*

$$v_j + \text{OPT}(p(j)) \geq \text{OPT}(j-1).$$

These facts form the first crucial component on which a dynamic programming solution is based: a recurrence equation that expresses the optimal solution (or its value) in terms of the optimal solutions to smaller subproblems.

Despite the simple reasoning that led to this point, (6.1) is already a significant development. It directly gives us a recursive algorithm to compute $\text{OPT}(n)$, assuming that we have already sorted the requests by finishing time and computed the values of $p(j)$ for each j.

```
Compute-Opt(j)
  If j = 0 then
    Return 0
  Else
    Return max(v_j+Compute-Opt(p(j)), Compute-Opt(j − 1))
  Endif
```

The correctness of the algorithm follows directly by induction on j:

(6.3) `Compute-Opt`(j) *correctly computes* $\text{OPT}(j)$ *for each* $j = 1, 2, \ldots, n$.

Proof. By definition $\text{OPT}(0) = 0$. Now, take some $j > 0$, and suppose by way of induction that `Compute-Opt`(i) correctly computes $\text{OPT}(i)$ for all $i < j$. By the induction hypothesis, we know that `Compute-Opt`$(p(j)) = \text{OPT}(p(j))$ and `Compute-Opt`$(j-1) = \text{OPT}(j-1)$; and hence from (6.1) it follows that

$$\begin{aligned}\text{OPT}(j) &= \max(v_j + \texttt{Compute-Opt}(p(j)), \texttt{Compute-Opt}(j-1)) \\ &= \texttt{Compute-Opt}(j). \quad \blacksquare\end{aligned}$$

Unfortunately, if we really implemented the algorithm `Compute-Opt` as just written, it would take exponential time to run in the worst case. For example, see Figure 6.3 for the tree of calls issued for the instance of Figure 6.2: the tree widens very quickly due to the recursive branching. To take a more extreme example, on a nicely layered instance like the one in Figure 6.4, where $p(j) = j - 2$ for each $j = 2, 3, 4, \ldots, n$, we see that `Compute-Opt`(j) generates separate recursive calls on problems of sizes $j-1$ and $j-2$. In other words, the total number of calls made to `Compute-Opt` on this instance will grow

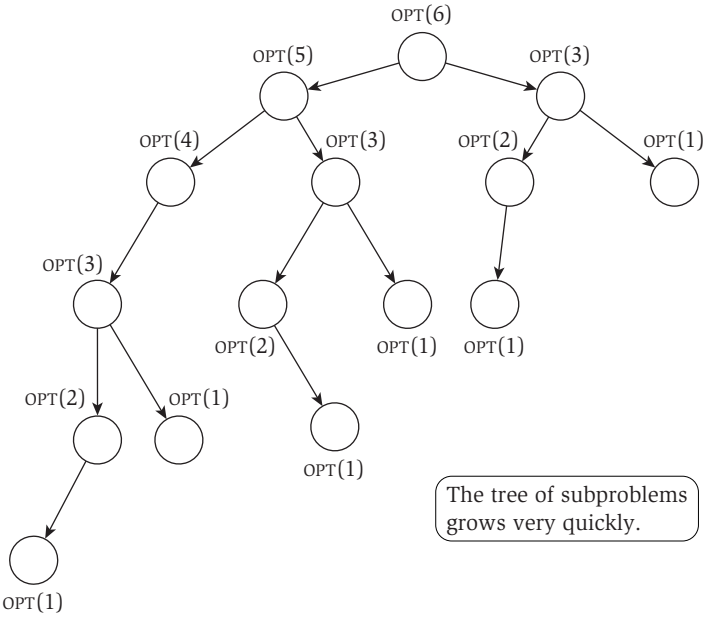


Figure 6.3 The tree of subproblems called by `Compute-Opt` on the problem instance of Figure 6.2.

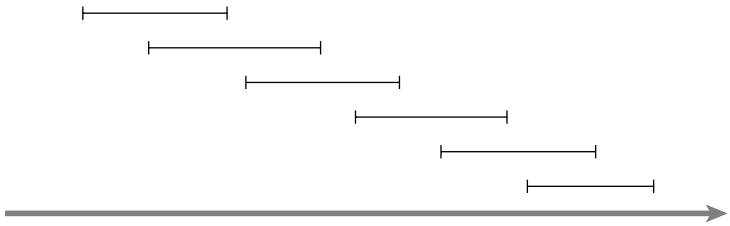


Figure 6.4 An instance of weighted interval scheduling on which the simple Compute-Opt recursion will take exponential time. The values of all intervals in this instance are 1.

like the Fibonacci numbers, which increase exponentially. Thus we have not achieved a polynomial-time solution.

Memoizing the Recursion

In fact, though, we're not so far from having a polynomial-time algorithm. A fundamental observation, which forms the second crucial component of a dynamic programming solution, is that our recursive algorithm Compute-Opt is really only solving $n+1$ different subproblems: Compute-Opt(0), Compute-Opt(1), ..., Compute-Opt(n). The fact that it runs in exponential time as written is simply due to the spectacular redundancy in the number of times it issues each of these calls.

How could we eliminate all this redundancy? We could store the value of Compute-Opt in a globally accessible place the first time we compute it and then simply use this precomputed value in place of all future recursive calls. This technique of saving values that have already been computed is referred to as *memoization*.

We implement the above strategy in the more "intelligent" procedure M-Compute-Opt. This procedure will make use of an array $M[0 \ldots n]$; $M[j]$ will start with the value "empty," but will hold the value of Compute-Opt(j) as soon as it is first determined. To determine OPT(n), we invoke M-Compute-Opt(n).

```
M-Compute-Opt(j)
  If j = 0 then
    Return 0
  Else if M[j] is not empty then
    Return M[j]
  Else
```

```
  Define M[j] = max(v_j+M-Compute-Opt(p(j)), M-Compute-Opt(j − 1))
   Return M[j]
 Endif
```

Analyzing the Memoized Version

Clearly, this looks very similar to our previous implementation of the algorithm; however, memoization has brought the running time way down.

(6.4) *The running time of* `M-Compute-Opt`(n) *is* $O(n)$ *(assuming the input intervals are sorted by their finish times).*

Proof. The time spent in a single call to `M-Compute-Opt` is $O(1)$, excluding the time spent in recursive calls it generates. So the running time is bounded by a constant times the number of calls ever issued to `M-Compute-Opt`. Since the implementation itself gives no explicit upper bound on this number of calls, we try to find a bound by looking for a good measure of "progress."

The most useful progress measure here is the number of entries in M that are not "empty." Initially this number is 0; but each time the procedure invokes the recurrence, issuing two recursive calls to `M-Compute-Opt`, it fills in a new entry, and hence increases the number of filled-in entries by 1. Since M has only $n + 1$ entries, it follows that there can be at most $O(n)$ calls to `M-Compute-Opt`, and hence the running time of `M-Compute-Opt`(n) is $O(n)$, as desired. ■

Computing a Solution in Addition to Its Value

So far we have simply computed the *value* of an optimal solution; presumably we want a full optimal set of intervals as well. It would be easy to extend `M-Compute-Opt` so as to keep track of an optimal solution in addition to its value: we could maintain an additional array S so that $S[i]$ contains an optimal set of intervals among $\{1, 2, \ldots, i\}$. Naively enhancing the code to maintain the solutions in the array S, however, would blow up the running time by an additional factor of $O(n)$: while a position in the M array can be updated in $O(1)$ time, writing down a set in the S array takes $O(n)$ time. We can avoid this $O(n)$ blow-up by not explicitly maintaining S, but rather by recovering the optimal solution from values saved in the array M after the optimum value has been computed.

We know from (6.2) that j belongs to an optimal solution for the set of intervals $\{1, \ldots, j\}$ if and only if $v_j + \text{OPT}(p(j)) \geq \text{OPT}(j - 1)$. Using this observation, we get the following simple procedure, which "traces back" through the array M to find the set of intervals in an optimal solution.

```
Find-Solution(j)
  If j = 0 then
    Output nothing
  Else
    If v_j + M[p(j)] ≥ M[j − 1] then
      Output j together with the result of Find-Solution(p(j))
    Else
      Output the result of Find-Solution(j − 1)
    Endif
  Endif
```

Since `Find-Solution` calls itself recursively only on strictly smaller values, it makes a total of $O(n)$ recursive calls; and since it spends constant time per call, we have

(6.5) *Given the array M of the optimal values of the sub-problems,* `Find-Solution` *returns an optimal solution in $O(n)$ time.*

6.2 Principles of Dynamic Programming: Memoization or Iteration over Subproblems

We now use the algorithm for the Weighted Interval Scheduling Problem developed in the previous section to summarize the basic principles of dynamic programming, and also to offer a different perspective that will be fundamental to the rest of the chapter: iterating over subproblems, rather than computing solutions recursively.

In the previous section, we developed a polynomial-time solution to the Weighted Interval Scheduling Problem by first designing an exponential-time recursive algorithm and then converting it (by memoization) to an efficient recursive algorithm that consulted a global array M of optimal solutions to subproblems. To really understand what is going on here, however, it helps to formulate an essentially equivalent version of the algorithm. It is this new formulation that most explicitly captures the essence of the dynamic programming technique, and it will serve as a general template for the algorithms we develop in later sections.

Designing the Algorithm

The key to the efficient algorithm is really the array M. It encodes the notion that we are using the value of optimal solutions to the subproblems on intervals $\{1, 2, \ldots, j\}$ for each j, and it uses (6.1) to define the value of $M[j]$ based on

values that come earlier in the array. Once we have the array M, the problem is solved: $M[n]$ contains the value of the optimal solution on the full instance, and `Find-Solution` can be used to trace back through M efficiently and return an optimal solution itself.

The point to realize, then, is that we can directly compute the entries in M by an iterative algorithm, rather than using memoized recursion. We just start with $M[0] = 0$ and keep incrementing j; each time we need to determine a value $M[j]$, the answer is provided by (6.1). The algorithm looks as follows.

```
Iterative-Compute-Opt
  M[0] = 0
  For j = 1, 2, ..., n
    M[j] = max(v_j + M[p(j)], M[j − 1])
  Endfor
```

Analyzing the Algorithm

By exact analogy with the proof of (6.3), we can prove by induction on j that this algorithm writes $\text{OPT}(j)$ in array entry $M[j]$; (6.1) provides the induction step. Also, as before, we can pass the filled-in array M to `Find-Solution` to get an optimal solution in addition to the value. Finally, the running time of `Iterative-Compute-Opt` is clearly $O(n)$, since it explicitly runs for n iterations and spends constant time in each.

An example of the execution of `Iterative-Compute-Opt` is depicted in Figure 6.5. In each iteration, the algorithm fills in one additional entry of the array M, by comparing the value of $v_j + M[p(j)]$ to the value of $M[j-1]$.

A Basic Outline of Dynamic Programming

This, then, provides a second efficient algorithm to solve the Weighted Interval Scheduling Problem. The two approaches clearly have a great deal of conceptual overlap, since they both grow from the insight contained in the recurrence (6.1). For the remainder of the chapter, we will develop dynamic programming algorithms using the second type of approach—iterative building up of subproblems—because the algorithms are often simpler to express this way. But in each case that we consider, there is an equivalent way to formulate the algorithm as a memoized recursion.

Most crucially, the bulk of our discussion about the particular problem of selecting intervals can be cast more generally as a rough template for designing dynamic programming algorithms. To set about developing an algorithm based on dynamic programming, one needs a collection of subproblems derived from the original problem that satisfies a few basic properties.

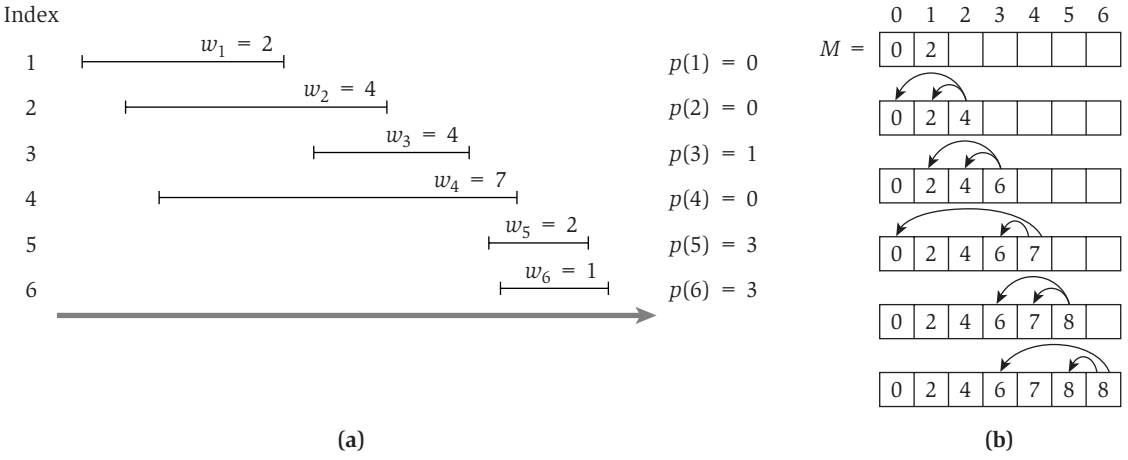


Figure 6.5 Part (b) shows the iterations of `Iterative-Compute-Opt` on the sample instance of Weighted Interval Scheduling depicted in part (a).

(i) There are only a polynomial number of subproblems.

(ii) The solution to the original problem can be easily computed from the solutions to the subproblems. (For example, the original problem may actually *be* one of the subproblems.)

(iii) There is a natural ordering on subproblems from "smallest" to "largest," together with an easy-to-compute recurrence (as in (6.1) and (6.2)) that allows one to determine the solution to a subproblem from the solutions to some number of smaller subproblems.

Naturally, these are informal guidelines. In particular, the notion of "smaller" in part (iii) will depend on the type of recurrence one has.

We will see that it is sometimes easier to start the process of designing such an algorithm by formulating a set of subproblems that looks natural, and then figuring out a recurrence that links them together; but often (as happened in the case of weighted interval scheduling), it can be useful to first define a recurrence by reasoning about the structure of an optimal solution, and then determine which subproblems will be necessary to unwind the recurrence. This chicken-and-egg relationship between subproblems and recurrences is a subtle issue underlying dynamic programming. It's never clear that a collection of subproblems will be useful until one finds a recurrence linking them together; but it can be difficult to think about recurrences in the absence of the "smaller" subproblems that they build on. In subsequent sections, we will develop further practice in managing this design trade-off.

6.3 Segmented Least Squares: Multi-way Choices

We now discuss a different type of problem, which illustrates a slightly more complicated style of dynamic programming. In the previous section, we developed a recurrence based on a fundamentally *binary* choice: either the interval n belonged to an optimal solution or it didn't. In the problem we consider here, the recurrence will involve what might be called "multi-way choices": at each step, we have a polynomial number of possibilities to consider for the structure of the optimal solution. As we'll see, the dynamic programming approach adapts to this more general situation very naturally.

As a separate issue, the problem developed in this section is also a nice illustration of how a clean algorithmic definition can formalize a notion that initially seems too fuzzy and nonintuitive to work with mathematically.

The Problem

Often when looking at scientific or statistical data, plotted on a two-dimensional set of axes, one tries to pass a "line of best fit" through the data, as in Figure 6.6.

This is a foundational problem in statistics and numerical analysis, formulated as follows. Suppose our data consists of a set P of n points in the plane, denoted $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$; and suppose $x_1 < x_2 < \cdots < x_n$. Given a line L defined by the equation $y = ax + b$, we say that the *error* of L with respect to P is the sum of its squared "distances" to the points in P:

$$\text{Error}(L, P) = \sum_{i=1}^{n} (y_i - ax_i - b)^2.$$

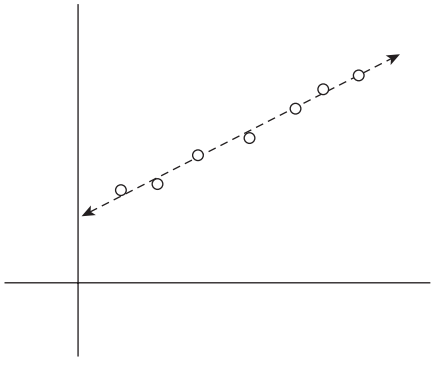


Figure 6.6 A "line of best fit."

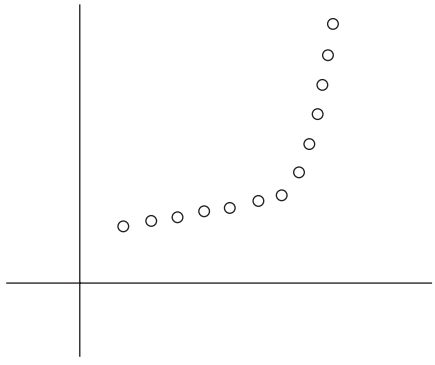


Figure 6.7 A set of points that lie approximately on two lines.

A natural goal is then to find the line with minimum error; this turns out to have a nice closed-form solution that can be easily derived using calculus. Skipping the derivation here, we simply state the result: The line of minimum error is $y = ax + b$, where

$$a = \frac{n \sum_i x_i y_i - \left(\sum_i x_i\right)\left(\sum_i y_i\right)}{n \sum_i x_i^2 - \left(\sum_i x_i\right)^2} \quad \text{and} \quad b = \frac{\sum_i y_i - a \sum_i x_i}{n}.$$

Now, here's a kind of issue that these formulas weren't designed to cover. Often we have data that looks something like the picture in Figure 6.7. In this case, we'd like to make a statement like: "The points lie roughly on a sequence of two lines." How could we formalize this concept?

Essentially, any single line through the points in the figure would have a terrible error; but if we use two lines, we could achieve quite a small error. So we could try formulating a new problem as follows: Rather than seek a single line of best fit, we are allowed to pass an arbitrary *set* of lines through the points, and we seek a set of lines that minimizes the error. But this fails as a good problem formulation, because it has a trivial solution: if we're allowed to fit the points with an arbitrarily large set of lines, we could fit the points perfectly by having a different line pass through each pair of consecutive points in P.

At the other extreme, we could try "hard-coding" the number two into the problem; we could seek the best fit using at most two lines. But this too misses a crucial feature of our intuition: We didn't start out with a preconceived idea that the points lay approximately on two lines; we concluded that from looking at the picture. For example, most people would say that the points in Figure 6.8 lie approximately on three lines.

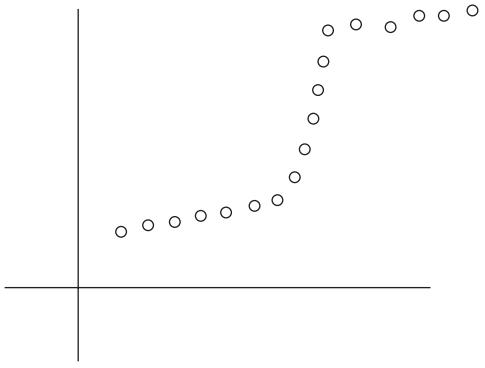


Figure 6.8 A set of points that lie approximately on three lines.

Thus, intuitively, we need a problem formulation that requires us to fit the points well, using as few lines as possible. We now formulate a problem—the *Segmented Least Squares Problem*—that captures these issues quite cleanly. The problem is a fundamental instance of an issue in data mining and statistics known as *change detection*: Given a sequence of data points, we want to identify a few points in the sequence at which a discrete *change* occurs (in this case, a change from one linear approximation to another).

Formulating the Problem As in the discussion above, we are given a set of points $P = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$, with $x_1 < x_2 < \cdots < x_n$. We will use p_i to denote the point (x_i, y_i). We must first partition P into some number of segments. Each *segment* is a subset of P that represents a contiguous set of x-coordinates; that is, it is a subset of the form $\{p_i, p_{i+1}, \ldots, p_{j-1}, p_j\}$ for some indices $i \leq j$. Then, for each segment S in our partition of P, we compute the line minimizing the error with respect to the points in S, according to the formulas above.

The *penalty* of a partition is defined to be a sum of the following terms.

(i) The number of segments into which we partition P, times a fixed, given multiplier $C > 0$.

(ii) For each segment, the error value of the optimal line through that segment.

Our goal in the Segmented Least Squares Problem is to find a partition of minimum penalty. This minimization captures the trade-offs we discussed earlier. We are allowed to consider partitions into any number of segments; as we increase the number of segments, we reduce the penalty terms in part (ii) of the definition, but we increase the term in part (i). (The multiplier C is provided

with the input, and by tuning C, we can penalize the use of additional lines to a greater or lesser extent.)

There are exponentially many possible partitions of P, and initially it is not clear that we should be able to find the optimal one efficiently. We now show how to use dynamic programming to find a partition of minimum penalty in time polynomial in n.

Designing the Algorithm

To begin with, we should recall the ingredients we need for a dynamic programming algorithm, as outlined at the end of Section 6.2.We want a polynomial number of subproblems, the solutions of which should yield a solution to the original problem; and we should be able to build up solutions to these subproblems using a recurrence. As with the Weighted Interval Scheduling Problem, it helps to think about some simple properties of the optimal solution. Note, however, that there is not really a direct analogy to weighted interval scheduling: there we were looking for a *subset* of n objects, whereas here we are seeking to *partition* n objects.

For segmented least squares, the following observation is very useful: The last point p_n belongs to a single segment in the optimal partition, and that segment begins at some earlier point p_i. This is the type of observation that can suggest the right set of subproblems: if we knew the identity of the *last* segment $p_i, \ldots, p_n$ (see Figure 6.9), then we could remove those points from consideration and recursively solve the problem on the remaining points $p_1, \ldots, p_{i-1}$.

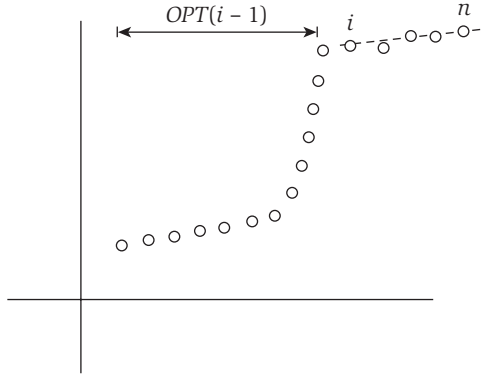


Figure 6.9 A possible solution: a single line segment fits points $p_i, p_{i+1}, \ldots, p_n$, and then an optimal solution is found for the remaining points $p_1, p_2, \ldots, p_{i-1}$.

Suppose we let $\text{OPT}(i)$ denote the optimum solution for the points $p_1, \ldots, p_i$, and we let $e_{i,j}$ denote the minimum error of any line with respect to $p_i, p_{i+1}, \ldots, p_j$. (We will write $\text{OPT}(0) = 0$ as a boundary case.) Then our observation above says the following.

(6.6) *If the last segment of the optimal partition is $p_i, \ldots, p_n$, then the value of the optimal solution is $\text{OPT}(n) = e_{i,n} + C + \text{OPT}(i-1)$.*

Using the same observation for the subproblem consisting of the points $p_1, \ldots, p_j$, we see that to get $\text{OPT}(j)$ we should find the best way to produce a final segment $p_i, \ldots, p_j$—paying the error plus an additive C for this segment—together with an optimal solution $\text{OPT}(i-1)$ for the remaining points. In other words, we have justified the following recurrence.

(6.7) *For the subproblem on the points $p_1, \ldots, p_j$,*

$$\text{OPT}(j) = \min_{1 \le i \le j} (e_{i,j} + C + \text{OPT}(i-1)),$$

and the segment $p_i, \ldots, p_j$ is used in an optimum solution for the subproblem if and only if the minimum is obtained using index i.

The hard part in designing the algorithm is now behind us. From here, we simply build up the solutions $\text{OPT}(i)$ in order of increasing i.

```
Segmented-Least-Squares(n)
  Array M[0 ... n]
  Set M[0] = 0
  For all pairs i ≤ j
    Compute the least squares error e_{i,j} for the segment p_i, ..., p_j
  Endfor
  For j = 1, 2, ..., n
    Use the recurrence (6.7) to compute M[j]
  Endfor
  Return M[n]
```

By analogy with the arguments for weighted interval scheduling, the correctness of this algorithm can be proved directly by induction, with (6.7) providing the induction step.

And as in our algorithm for weighted interval scheduling, we can trace back through the array M to compute an optimum partition.

```
Find-Segments(j)
  If j = 0 then
    Output nothing
  Else
    Find an i that minimizes e_{i,j} + C + M[i − 1]
    Output the segment {p_i, ..., p_j} and the result of
           Find-Segments(i − 1)
  Endif
```

Analyzing the Algorithm

Finally, we consider the running time of `Segmented-Least-Squares`. First we need to compute the values of all the least-squares errors $e_{i,j}$. To perform a simple accounting of the running time for this, we note that there are $O(n^2)$ pairs (i, j) for which this computation is needed; and for each pair (i, j), we can use the formula given at the beginning of this section to compute $e_{i,j}$ in $O(n)$ time. Thus the total running time to compute all $e_{i,j}$ values is $O(n^3)$.

Following this, the algorithm has n iterations, for values $j = 1, \ldots, n$. For each value of j, we have to determine the minimum in the recurrence (6.7) to fill in the array entry $M[j]$; this takes time $O(n)$ for each j, for a total of $O(n^2)$. Thus the running time is $O(n^2)$ once all the $e_{i,j}$ values have been determined.[1]

6.4 Subset Sums and Knapsacks: Adding a Variable

We're seeing more and more that issues in scheduling provide a rich source of practically motivated algorithmic problems. So far we've considered problems in which requests are specified by a given interval of time on a resource, as well as problems in which requests have a duration and a deadline but do not mandate a particular interval during which they need to be done.

In this section, we consider a version of the second type of problem, with durations and deadlines, which is difficult to solve directly using the techniques we've seen so far. We will use dynamic programming to solve the problem, but with a twist: the "obvious" set of subproblems will turn out not to be enough, and so we end up creating a richer collection of subproblems. As

[1] In this analysis, the running time is dominated by the $O(n^3)$ needed to compute all $e_{i,j}$ values. But, in fact, it is possible to compute all these values in $O(n^2)$ time, which brings the running time of the full algorithm down to $O(n^2)$. The idea, whose details we will leave as an exercise for the reader, is to first compute $e_{i,j}$ for all pairs (i, j) where $j - i = 1$, then for all pairs where $j - i = 2$, then $j - i = 3$, and so forth. This way, when we get to a particular $e_{i,j}$ value, we can use the ingredients of the calculation for $e_{i,j-1}$ to determine $e_{i,j}$ in constant time.

we will see, this is done by adding a new variable to the recurrence underlying the dynamic program.

The Problem

In the scheduling problem we consider here, we have a single machine that can process jobs, and we have a set of requests $\{1, 2, \ldots, n\}$. We are only able to use this resource for the period between time 0 and time W, for some number W. Each request corresponds to a job that requires time w_i to process. If our goal is to process jobs so as to keep the machine as busy as possible up to the "cut-off" W, which jobs should we choose?

More formally, we are given n items $\{1, \ldots, n\}$, and each has a given nonnegative weight w_i (for $i = 1, \ldots, n$). We are also given a bound W. We would like to select a subset S of the items so that $\sum_{i \in S} w_i \leq W$ and, subject to this restriction, $\sum_{i \in S} w_i$ is as large as possible. We will call this the *Subset Sum Problem*.

This problem is a natural special case of a more general problem called the *Knapsack Problem*, where each request i has both a *value* v_i and a *weight* w_i. The goal in this more general problem is to select a subset of maximum total value, subject to the restriction that its total weight not exceed W. Knapsack problems often show up as subproblems in other, more complex problems. The name *knapsack* refers to the problem of filling a knapsack of capacity W as full as possible (or packing in as much value as possible), using a subset of the items $\{1, \ldots, n\}$. We will use *weight* or *time* when referring to the quantities w_i and W.

Since this resembles other scheduling problems we've seen before, it's natural to ask whether a greedy algorithm can find the optimal solution. It appears that the answer is no—at least, no efficient greedy rule is known that always constructs an optimal solution. One natural greedy approach to try would be to sort the items by decreasing weight—or at least to do this for all items of weight at most W—and then start selecting items in this order as long as the total weight remains below W. But if W is a multiple of 2, and we have three items with weights $\{W/2 + 1, W/2, W/2\}$, then we see that this greedy algorithm will not produce the optimal solution. Alternately, we could sort by *increasing* weight and then do the same thing; but this fails on inputs like $\{1, W/2, W/2\}$.

The goal of this section is to show how to use dynamic programming to solve this problem. Recall the main principles of dynamic programming: We have to come up with a small number of subproblems so that each subproblem can be solved easily from "smaller" subproblems, and the solution to the original problem can be obtained easily once we know the solutions to all

the subproblems. The tricky issue here lies in figuring out a good set of subproblems.

Designing the Algorithm

A False Start One general strategy, which worked for us in the case of Weighted Interval Scheduling, is to consider subproblems involving only the first i requests. We start by trying this strategy here. We use the notation $\text{OPT}(i)$, analogously to the notation used before, to denote the best possible solution using a subset of the requests $\{1, \ldots, i\}$. The key to our method for the Weighted Interval Scheduling Problem was to concentrate on an optimal solution $\mathcal{O}$ to our problem and consider two cases, depending on whether or not the last request n is accepted or rejected by this optimum solution. Just as in that case, we have the first part, which follows immediately from the definition of $\text{OPT}(i)$.

- If $n \notin \mathcal{O}$, then $\text{OPT}(n) = \text{OPT}(n-1)$.

Next we have to consider the case in which $n \in \mathcal{O}$. What we'd like here is a simple recursion, which tells us the best possible value we can get for solutions that contain the last request n. For Weighted Interval Scheduling this was easy, as we could simply delete each request that conflicted with request n. In the current problem, this is not so simple. Accepting request n does not immediately imply that we have to reject any other request. Instead, it means that for the subset of requests $S \subseteq \{1, \ldots, n-1\}$ that we will accept, we have less available weight left: a weight of w_n is used on the accepted request n, and we only have $W - w_n$ weight left for the set S of remaining requests that we accept. See Figure 6.10.

A Better Solution This suggests that we need more subproblems: To find out the value for $\text{OPT}(n)$ we not only need the value of $\text{OPT}(n-1)$, but we also need to know the best solution we can get using a subset of the first $n-1$ items and total allowed weight $W - w_n$. We are therefore going to use many more subproblems: one for each initial set $\{1, \ldots, i\}$ of the items, and each possible

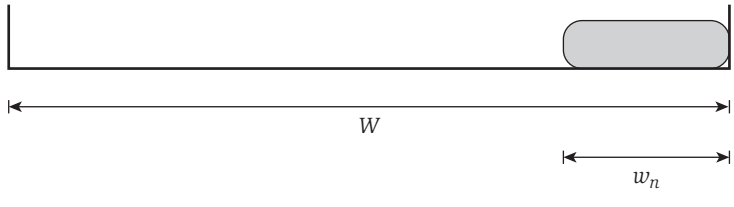


Figure 6.10 After item n is included in the solution, a weight of w_n is used up and there is $W - w_n$ available weight left.

value for the remaining available weight w. Assume that W is an integer, and all requests $i = 1, \ldots, n$ have integer weights w_i. We will have a subproblem for each $i = 0, 1, \ldots, n$ and each integer $0 \leq w \leq W$. We will use $\text{OPT}(i, w)$ to denote the value of the optimal solution using a subset of the items $\{1, \ldots, i\}$ with maximum allowed weight w, that is,

$$\text{OPT}(i, w) = \max_S \sum_{j \in S} w_j,$$

where the maximum is over subsets $S \subseteq \{1, \ldots, i\}$ that satisfy $\sum_{j \in S} w_j \leq w$. Using this new set of subproblems, we will be able to express the value $\text{OPT}(i, w)$ as a simple expression in terms of values from smaller problems. Moreover, $\text{OPT}(n, W)$ is the quantity we're looking for in the end. As before, let $\mathcal{O}$ denote an optimum solution for the original problem.

- If $n \notin \mathcal{O}$, then $\text{OPT}(n, W) = \text{OPT}(n-1, W)$, since we can simply ignore item n.
- If $n \in \mathcal{O}$, then $\text{OPT}(n, W) = w_n + \text{OPT}(n-1, W - w_n)$, since we now seek to use the remaining capacity of $W - w_n$ in an optimal way across items $1, 2, \ldots, n-1$.

When the n^{th} item is too big, that is, $W < w_n$, then we must have $\text{OPT}(n, W) = \text{OPT}(n-1, W)$. Otherwise, we get the optimum solution allowing all n requests by taking the better of these two options. Using the same line of argument for the subproblem for items $\{1, \ldots, i\}$, and maximum allowed weight w, gives us the following recurrence.

(6.8) *If $w < w_i$ then $\text{OPT}(i, w) = \text{OPT}(i-1, w)$. Otherwise*

$$\text{OPT}(i, w) = \max(\text{OPT}(i-1, w), w_i + \text{OPT}(i-1, w - w_i)).$$

As before, we want to design an algorithm that builds up a table of all $\text{OPT}(i, w)$ values while computing each of them at most once.

```
Subset-Sum(n, W)
  Array M[0...n, 0...W]
  Initialize M[0, w] = 0 for each w = 0, 1, ..., W
  For i = 1, 2, ..., n
     For w = 0, ..., W
        Use the recurrence (6.8) to compute M[i, w]
     Endfor
  Endfor
  Return M[n, W]
```

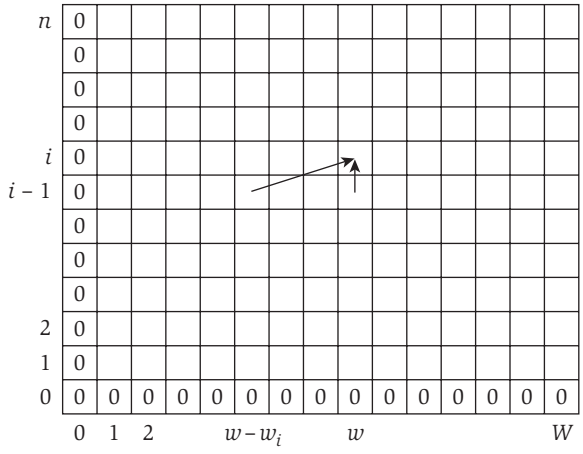


Figure 6.11 The two-dimensional table of OPT values. The leftmost column and bottom row is always 0. The entry for $\text{OPT}(i, w)$ is computed from the two other entries $\text{OPT}(i-1, w)$ and $\text{OPT}(i-1, w-w_i)$, as indicated by the arrows.

Using (6.8) one can immediately prove by induction that the returned value $M[n, W]$ is the optimum solution value for the requests $1, \ldots, n$ and available weight W.

Analyzing the Algorithm

Recall the tabular picture we considered in Figure 6.5, associated with weighted interval scheduling, where we also showed the way in which the array M for that algorithm was iteratively filled in. For the algorithm we've just designed, we can use a similar representation, but we need a two-dimensional table, reflecting the two-dimensional array of subproblems that is being built up. Figure 6.11 shows the building up of subproblems in this case: the value $M[i, w]$ is computed from the two other values $M[i-1, w]$ and $M[i-1, w-w_i]$.

As an example of this algorithm executing, consider an instance with weight limit $W = 6$, and $n = 3$ items of sizes $w_1 = w_2 = 2$ and $w_3 = 3$. We find that the optimal value $\text{OPT}(3, 6) = 5$ (which we get by using the third item and one of the first two items). Figure 6.12 illustrates the way the algorithm fills in the two-dimensional table of OPT values row by row.

Next we will worry about the running time of this algorithm. As before in the case of weighted interval scheduling, we are building up a table of solutions M, and we compute each of the values $M[i, w]$ in $O(1)$ time using the previous values. Thus the running time is proportional to the number of entries in the table.

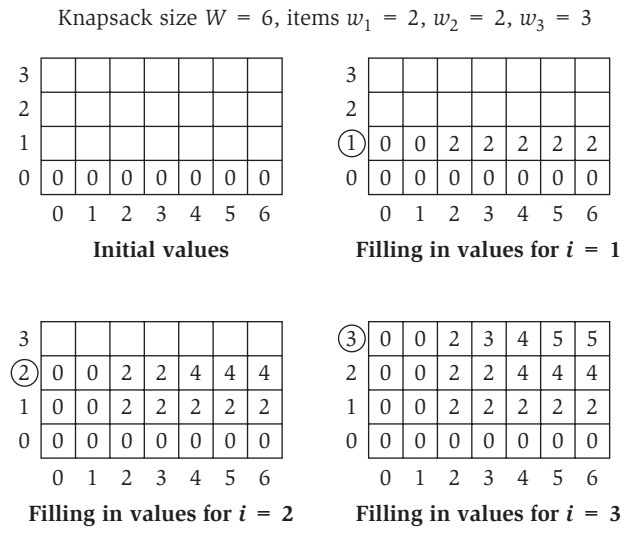


Figure 6.12 The iterations of the algorithm on a sample instance of the Subset Sum Problem.

(6.9) *The* `Subset-Sum`(n, W) *Algorithm correctly computes the optimal value of the problem, and runs in* $O(nW)$ *time.*

Note that this method is not as efficient as our dynamic program for the Weighted Interval Scheduling Problem. Indeed, its running time is not a polynomial function of n; rather, it is a polynomial function of n and W, the largest integer involved in defining the problem. We call such algorithms *pseudo-polynomial*. Pseudo-polynomial algorithms can be reasonably efficient when the numbers $\{w_i\}$ involved in the input are reasonably small; however, they become less practical as these numbers grow large.

To recover an optimal set S of items, we can trace back through the array M by a procedure similar to those we developed in the previous sections.

(6.10) *Given a table* M *of the optimal values of the subproblems, the optimal set* S *can be found in* $O(n)$ *time.*

Extension: The Knapsack Problem

The Knapsack Problem is a bit more complex than the scheduling problem we discussed earlier. Consider a situation in which each item i has a nonnegative weight w_i as before, and also a distinct *value* v_i. Our goal is now to find a

subset S of maximum value $\sum_{i \in S} v_i$, subject to the restriction that the total weight of the set should not exceed W: $\sum_{i \in S} w_i \leq W$.

It is not hard to extend our dynamic programming algorithm to this more general problem. We use the analogous set of subproblems, $\text{OPT}(i, w)$, to denote the value of the optimal solution using a subset of the items $\{1, \ldots, i\}$ and maximum available weight w. We consider an optimal solution $\mathcal{O}$, and identify two cases depending on whether or not $n \in \mathcal{O}$.

- If $n \notin \mathcal{O}$, then $\text{OPT}(n, W) = \text{OPT}(n-1, W)$.
- If $n \in \mathcal{O}$, then $\text{OPT}(n, W) = v_n + \text{OPT}(n-1, W - w_n)$.

Using this line of argument for the subproblems implies the following analogue of (6.8).

(6.11) *If $w < w_i$ then $\text{OPT}(i, w) = \text{OPT}(i-1, w)$. Otherwise*

$$\text{OPT}(i, w) = \max(\text{OPT}(i-1, w), v_i + \text{OPT}(i-1, w - w_i)).$$

Using this recurrence, we can write down a completely analogous dynamic programming algorithm, and this implies the following fact.

(6.12) *The Knapsack Problem can be solved in $O(nW)$ time.*

6.5 RNA Secondary Structure: Dynamic Programming over Intervals

In the Knapsack Problem, we were able to formulate a dynamic programming algorithm by adding a new variable. A different but very common way by which one ends up adding a variable to a dynamic program is through the following scenario. We start by thinking about the set of subproblems on $\{1, 2, \ldots, j\}$, for all choices of j, and find ourselves unable to come up with a natural recurrence. We then look at the larger set of subproblems on $\{i, i+1, \ldots, j\}$ for all choices of i and j (where $i \leq j$), and find a natural recurrence relation on these subproblems. In this way, we have added the second variable i; the effect is to consider a subproblem for every contiguous *interval* in $\{1, 2, \ldots, n\}$.

There are a few canonical problems that fit this profile; those of you who have studied parsing algorithms for context-free grammars have probably seen at least one dynamic programming algorithm in this style. Here we focus on the problem of RNA secondary structure prediction, a fundamental issue in computational biology.

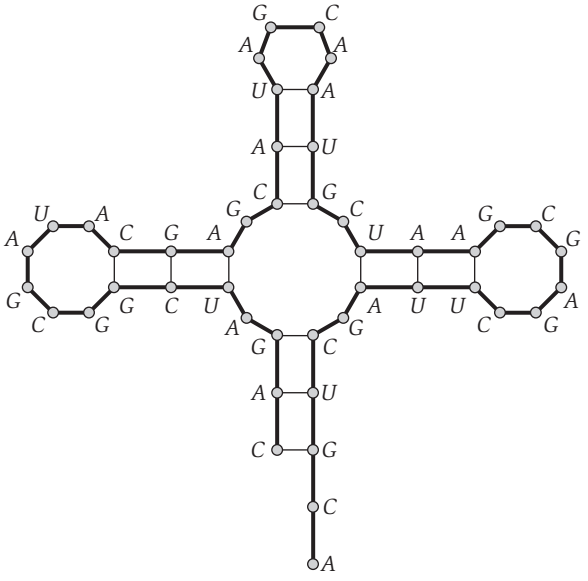


Figure 6.13 An RNA secondary structure. Thick lines connect adjacent elements of the sequence; thin lines indicate pairs of elements that are matched.

The Problem

As one learns in introductory biology classes, Watson and Crick posited that double-stranded DNA is "zipped" together by complementary base-pairing. Each strand of DNA can be viewed as a string of *bases*, where each base is drawn from the set $\{A, C, G, T\}$.[2] The bases A and T pair with each other, and the bases C and G pair with each other; it is these A-T and C-G pairings that hold the two strands together.

Now, single-stranded RNA molecules are key components in many of the processes that go on inside a cell, and they follow more or less the same structural principles. However, unlike double-stranded DNA, there's no "second strand" for the RNA to stick to; so it tends to loop back and form base pairs with itself, resulting in interesting shapes like the one depicted in Figure 6.13. The set of pairs (and resulting shape) formed by the RNA molecule through this process is called the *secondary structure*, and understanding the secondary structure is essential for understanding the behavior of the molecule.

[2] Adenine, cytosine, guanine, and thymine, the four basic units of DNA.

For our purposes, a single-stranded RNA molecule can be viewed as a sequence of n symbols (bases) drawn from the alphabet $\{A, C, G, U\}$.[3] Let $B = b_1 b_2 \cdots b_n$ be a single-stranded RNA molecule, where each $b_i \in \{A, C, G, U\}$. To a first approximation, one can model its secondary structure as follows. As usual, we require that A pairs with U, and C pairs with G; we also require that each base can pair with at most one other base—in other words, the set of base pairs forms a *matching*. It also turns out that secondary structures are (again, to a first approximation) "knot-free," which we will formalize as a kind of *noncrossing* condition below.

Thus, concretely, we say that a *secondary structure on* B is a set of pairs $S = \{(i, j)\}$, where $i, j \in \{1, 2, \ldots, n\}$, that satisfies the following conditions.

(i) *(No sharp turns.)* The ends of each pair in S are separated by at least four intervening bases; that is, if $(i, j) \in S$, then $i < j - 4$.

(ii) The elements of any pair in S consist of either $\{A, U\}$ or $\{C, G\}$ (in either order).

(iii) S is a matching: no base appears in more than one pair.

(iv) *(The noncrossing condition.)* If (i, j) and (k, ℓ) are two pairs in S, then we cannot have $i < k < j < \ell$. (See Figure 6.14 for an illustration.)

Note that the RNA secondary structure in Figure 6.13 satisfies properties (i) through (iv). From a structural point of view, condition (i) arises simply because the RNA molecule cannot bend too sharply; and conditions (ii) and (iii) are the fundamental Watson-Crick rules of base-pairing. Condition (iv) is the striking one, since it's not obvious why it should hold in nature. But while there are sporadic exceptions to it in real molecules (via so-called *pseudoknotting*), it does turn out to be a good approximation to the spatial constraints on real RNA secondary structures.

Now, out of all the secondary structures that are possible for a single RNA molecule, which are the ones that are likely to arise under physiological conditions? The usual hypothesis is that a single-stranded RNA molecule will form the secondary structure with the optimum total free energy. The correct model for the free energy of a secondary structure is a subject of much debate; but a first approximation here is to assume that the free energy of a secondary structure is proportional simply to the *number* of base pairs that it contains.

Thus, having said all this, we can state the basic RNA secondary structure prediction problem very simply: We want an efficient algorithm that takes

[3] Note that the symbol T from the alphabet of DNA has been replaced by a U, but this is not important for us here.

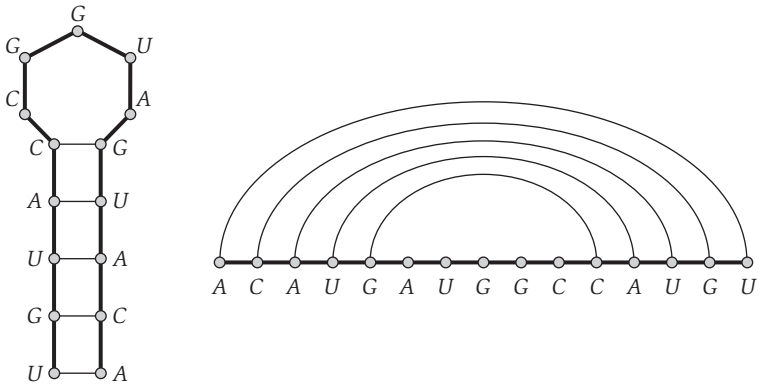


Figure 6.14 Two views of an RNA secondary structure. In the second view, (b), the string has been "stretched" lengthwise, and edges connecting matched pairs appear as noncrossing "bubbles" over the string.

a single-stranded RNA molecule $B = b_1 b_2 \cdots b_n$ and determines a secondary structure S with the maximum possible number of base pairs.

Designing and Analyzing the Algorithm

A First Attempt at Dynamic Programming The natural first attempt to apply dynamic programming would presumably be based on the following subproblems: We say that OPT(j) is the maximum number of base pairs in a secondary structure on $b_1 b_2 \cdots b_j$. By the no-sharp-turns condition above, we know that OPT(j) $= 0$ for $j \leq 5$; and we know that OPT(n) is the solution we're looking for.

The trouble comes when we try writing down a recurrence that expresses OPT(j) in terms of the solutions to smaller subproblems. We can get partway there: in the optimal secondary structure on $b_1 b_2 \cdots b_j$, it's the case that either

- j is not involved in a pair; or
- j pairs with t for some $t < j - 4$.

In the first case, we just need to consult our solution for OPT($j - 1$). The second case is depicted in Figure 6.15(a); because of the noncrossing condition, we now know that no pair can have one end between 1 and $t - 1$ and the other end between $t + 1$ and $j - 1$. We've therefore effectively isolated two new subproblems: one on the bases $b_1 b_2 \cdots b_{t-1}$, and the other on the bases $b_{t+1} \cdots b_{j-1}$. The first is solved by OPT($t - 1$), but the second is not on our list of subproblems, because it does not begin with b_1.

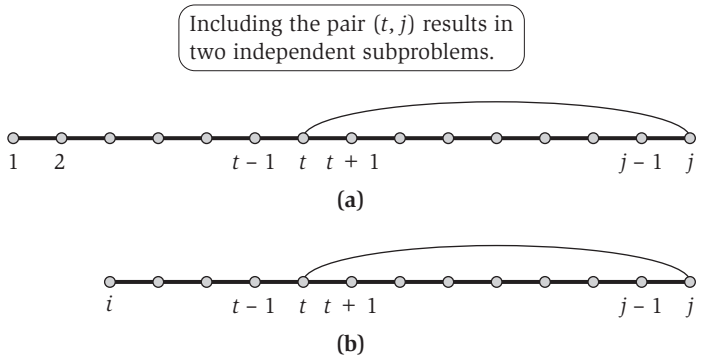


Figure 6.15 Schematic views of the dynamic programming recurrence using (a) one variable, and (b) two variables.

This is the insight that makes us realize we need to add a variable. We need to be able to work with subproblems that do not begin with b_1; in other words, we need to consider subproblems on $b_i b_{i+1} \cdots b_j$ for all choices of $i \leq j$.

Dynamic Programming over Intervals Once we make this decision, our previous reasoning leads straight to a successful recurrence. Let $\text{OPT}(i, j)$ denote the maximum number of base pairs in a secondary structure on $b_i b_{i+1} \cdots b_j$. The no-sharp-turns condition lets us initialize $\text{OPT}(i, j) = 0$ whenever $i \geq j - 4$. (For notational convenience, we will also allow ourselves to refer to $\text{OPT}(i, j)$ even when $i > j$; in this case, its value is 0.)

Now, in the optimal secondary structure on $b_i b_{i+1} \cdots b_j$, we have the same alternatives as before:

- j is not involved in a pair; or
- j pairs with t for some $t < j - 4$.

In the first case, we have $\text{OPT}(i, j) = \text{OPT}(i, j - 1)$. In the second case, depicted in Figure 6.15(b), we recur on the two subproblems $\text{OPT}(i, t - 1)$ and $\text{OPT}(t + 1, j - 1)$; as argued above, the noncrossing condition has isolated these two subproblems from each other.

We have therefore justified the following recurrence.

(6.13) $\text{OPT}(i, j) = \max(\text{OPT}(i, j - 1), \max(1 + \text{OPT}(i, t - 1) + \text{OPT}(t + 1, j - 1)))$, *where the* max *is taken over* t *such that* b_t *and* b_j *are an allowable base pair (under conditions (i) and (ii) from the definition of a secondary structure).*

Now we just have to make sure we understand the proper order in which to build up the solutions to the subproblems. The form of (6.13) reveals that we're always invoking the solution to subproblems on *shorter* intervals: those

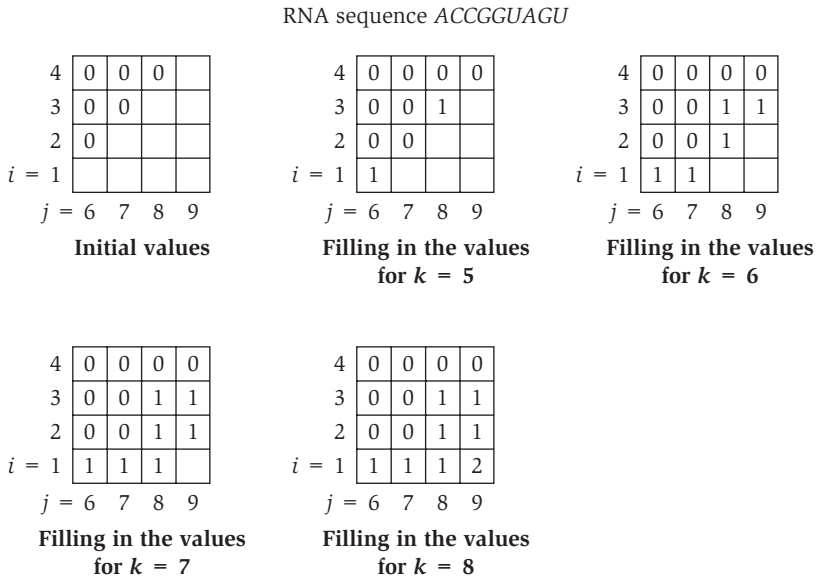


Figure 6.16 The iterations of the algorithm on a sample instance of the RNA Secondary Structure Prediction Problem.

for which $k = j - i$ is smaller. Thus things will work without any trouble if we build up the solutions in order of increasing interval length.

```
Initialize OPT(i, j) = 0 whenever i ≥ j − 4
For k = 5, 6, ..., n − 1
  For i = 1, 2, ... n − k
    Set j = i + k
    Compute OPT(i, j) using the recurrence in (6.13)
  Endfor
Endfor
Return OPT(1, n)
```

As an example of this algorithm executing, we consider the input *ACCGGUAGU*, a subsequence of the sequence in Figure 6.14. As with the Knapsack Problem, we need two dimensions to depict the array M: one for the left endpoint of the interval being considered, and one for the right endpoint. In the figure, we only show entries corresponding to $[i, j]$ pairs with $i < j - 4$, since these are the only ones that can possibly be nonzero.

It is easy to bound the running time: there are $O(n^2)$ subproblems to solve, and evaluating the recurrence in (6.13) takes time $O(n)$ for each. Thus the running time is $O(n^3)$.

As always, we can recover the secondary structure itself (not just its value) by recording how the minima in (6.13) are achieved and tracing back through the computation.

6.6 Sequence Alignment

For the remainder of this chapter, we consider two further dynamic programming algorithms that each have a wide range of applications. In the next two sections we discuss *sequence alignment*, a fundamental problem that arises in comparing strings. Following this, we turn to the problem of computing shortest paths in graphs when edges have costs that may be negative.

The Problem

Dictionaries on the Web seem to get more and more useful: often it seems easier to pull up a bookmarked online dictionary than to get a physical dictionary down from the bookshelf. And many online dictionaries offer functions that you can't get from a printed one: if you're looking for a definition and type in a word it doesn't contain—say, *ocurrance*—it will come back and ask, "Perhaps you mean *occurrence*?" How does it do this? Did it truly know what you had in mind?

Let's defer the second question to a different book and think a little about the first one. To decide what you probably meant, it would be natural to search the dictionary for the word most "similar" to the one you typed in. To do this, we have to answer the question: How should we define similarity between two words or strings?

Intuitively, we'd like to say that *ocurrance* and *occurrence* are similar because we can make the two words identical if we add a *c* to the first word and change the *a* to an *e*. Since neither of these changes seems so large, we conclude that the words are quite similar. To put it another way, we can *nearly* line up the two words letter by letter:

```
o-currance
occurrence
```

The hyphen (-) indicates a *gap* where we had to add a letter to the second word to get it to line up with the first. Moreover, our lining up is not perfect in that an *e* is lined up with an *a*.

We want a model in which similarity is determined roughly by the number of gaps and mismatches we incur when we line up the two words. Of course, there are many possible ways to line up the two words; for example, we could have written

```
o-curr-ance
occurre-nce
```

which involves three gaps and no mismatches. Which is better: one gap and one mismatch, or three gaps and no mismatches?

This discussion has been made easier because we know roughly what the correspondence ought to look like. When the two strings don't look like English words—for example, abbbaabbbbaab and ababaaabbbbbab—it may take a little work to decide whether they can be lined up nicely or not:

```
abbbaa--bbbbaab
ababaaabbbbba-b
```

Dictionary interfaces and spell-checkers are not the most computationally intensive application for this type of problem. In fact, determining similarities among strings is one of the central computational problems facing molecular biologists today.

Strings arise very naturally in biology: an organism's *genome*—its full set of genetic material—is divided up into giant linear DNA molecules known as *chromosomes,* each of which serves conceptually as a one-dimensional chemical storage device. Indeed, it does not obscure reality very much to think of it as an enormous linear *tape*, containing a string over the alphabet $\{A, C, G, T\}$. The string of symbols encodes the instructions for building protein molecules; using a chemical mechanism for reading portions of the chromosome, a cell can construct proteins that in turn control its metabolism.

Why is similarity important in this picture? To a first approximation, the sequence of symbols in an organism's genome can be viewed as determining the properties of the organism. So suppose we have two strains of bacteria, X and Y, which are closely related evolutionarily. Suppose further that we've determined that a certain substring in the DNA of X codes for a certain kind of toxin. Then, if we discover a very "similar" substring in the DNA of Y, we might be able to hypothesize, before performing any experiments at all, that this portion of the DNA in Y codes for a similar kind of toxin. This use of computation to guide decisions about biological experiments is one of the hallmarks of the field of *computational biology*.

All this leaves us with the same question we asked initially, while typing badly spelled words into our online dictionary: How should we define the notion of *similarity* between two strings?

In the early 1970s, the two molecular biologists Needleman and Wunsch proposed a definition of similarity, which, basically unchanged, has become

the standard definition in use today. Its position as a standard was reinforced by its simplicity and intuitive appeal, as well as through its independent discovery by several other researchers around the same time. Moreover, this definition of similarity came with an efficient dynamic programming algorithm to compute it. In this way, the paradigm of dynamic programming was independently discovered by biologists some twenty years after mathematicians and computer scientists first articulated it.

The definition is motivated by the considerations we discussed above, and in particular by the notion of "lining up" two strings. Suppose we are given two strings X and Y, where X consists of the sequence of symbols $x_1x_2 \cdots x_m$ and Y consists of the sequence of symbols $y_1y_2 \cdots y_n$. Consider the sets $\{1, 2, \ldots, m\}$ and $\{1, 2, \ldots, n\}$ as representing the different positions in the strings X and Y, and consider a matching of these sets; recall that a *matching* is a set of ordered pairs with the property that each item occurs in at most one pair. We say that a matching M of these two sets is an *alignment* if there are no "crossing" pairs: if $(i, j), (i', j') \in M$ and $i < i'$, then $j < j'$. Intuitively, an alignment gives a way of lining up the two strings, by telling us which pairs of positions will be lined up with one another. Thus, for example,

```
stop-
-tops
```

corresponds to the alignment $\{(2, 1), (3, 2), (4, 3)\}$.

Our definition of similarity will be based on finding the *optimal* alignment between X and Y, according to the following criteria. Suppose M is a given alignment between X and Y.

- First, there is a parameter $\delta > 0$ that defines a *gap penalty*. For each position of X or Y that is not matched in M—it is a *gap*—we incur a cost of δ.
- Second, for each pair of letters p, q in our alphabet, there is a *mismatch cost* of α_{pq} for lining up p with q. Thus, for each $(i, j) \in M$, we pay the appropriate mismatch cost $\alpha_{x_i y_j}$ for lining up x_i with y_j. One generally assumes that $\alpha_{pp} = 0$ for each letter p—there is no mismatch cost to line up a letter with another copy of itself—although this will not be necessary in anything that follows.
- The *cost* of M is the sum of its gap and mismatch costs, and we seek an alignment of minimum cost.

The process of minimizing this cost is often referred to as *sequence alignment* in the biology literature. The quantities δ and $\{\alpha_{pq}\}$ are external parameters that must be plugged into software for sequence alignment; indeed, a lot of work goes into choosing the settings for these parameters. From our point of

view, in designing an algorithm for sequence alignment, we will take them as given. To go back to our first example, notice how these parameters determine which alignment of *ocurrance* and *occurrence* we should prefer: the first is strictly better if and only if $\delta + \alpha_{ae} < 3\delta$.

Designing the Algorithm

We now have a concrete numerical definition for the similarity between strings X and Y: it is the minimum cost of an alignment between X and Y. The lower this cost, the more similar we declare the strings to be. We now turn to the problem of computing this minimum cost, and an optimal alignment that yields it, for a given pair of strings X and Y.

One of the approaches we could try for this problem is dynamic programming, and we are motivated by the following basic dichotomy.

- In the optimal alignment M, either $(m, n) \in M$ or $(m, n) \notin M$. (That is, either the last symbols in the two strings are matched to each other, or they aren't.)

By itself, this fact would be too weak to provide us with a dynamic programming solution. Suppose, however, that we compound it with the following basic fact.

(6.14) *Let M be any alignment of X and Y. If $(m, n) \notin M$, then either the m^{th} position of X or the n^{th} position of Y is not matched in M.*

Proof. Suppose by way of contradiction that $(m, n) \notin M$, and there are numbers $i < m$ and $j < n$ so that $(m, j) \in M$ and $(i, n) \in M$. But this contradicts our definition of *alignment*: we have $(i, n), (m, j) \in M$ with $i < m$, but $n > i$ so the pairs (i, n) and (m, j) cross. ■

There is an equivalent way to write (6.14) that exposes three alternative possibilities, and leads directly to the formulation of a recurrence.

(6.15) *In an optimal alignment M, at least one of the following is true:*

(i) $(m, n) \in M$*; or*

(ii) the m^{th} position of X is not matched; or

(iii) the n^{th} position of Y is not matched.

Now, let $\text{OPT}(i, j)$ denote the minimum cost of an alignment between $x_1x_2 \cdots x_i$ and $y_1y_2 \cdots y_j$. If case (i) of (6.15) holds, we pay $\alpha_{x_m y_n}$ and then align $x_1x_2 \cdots x_{m-1}$ as well as possible with $y_1y_2 \cdots y_{n-1}$; we get $\text{OPT}(m, n) = \alpha_{x_m y_n} + \text{OPT}(m-1, n-1)$. If case (ii) holds, we pay a gap cost of δ since the m^{th} position of X is not matched, and then we align $x_1x_2 \cdots x_{m-1}$ as well as

possible with $y_1 y_2 \cdots y_n$. In this way, we get $\text{OPT}(m, n) = \delta + \text{OPT}(m-1, n)$. Similarly, if case (iii) holds, we get $\text{OPT}(m, n) = \delta + \text{OPT}(m, n-1)$.

Using the same argument for the subproblem of finding the minimum-cost alignment between $x_1 x_2 \cdots x_i$ and $y_1 y_2 \cdots y_j$, we get the following fact.

(6.16) *The minimum alignment costs satisfy the following recurrence for $i \geq 1$ and $j \geq 1$:*

$$\text{OPT}(i, j) = \min[\alpha_{x_i y_j} + \text{OPT}(i-1, j-1), \delta + \text{OPT}(i-1, j), \delta + \text{OPT}(i, j-1)].$$

Moreover, (i, j) is in an optimal alignment M for this subproblem if and only if the minimum is achieved by the first of these values.

We have maneuvered ourselves into a position where the dynamic programming algorithm has become clear: We build up the values of $\text{OPT}(i, j)$ using the recurrence in (6.16). There are only $O(mn)$ subproblems, and $\text{OPT}(m, n)$ is the value we are seeking.

We now specify the algorithm to compute the value of the optimal alignment. For purposes of initialization, we note that $\text{OPT}(i, 0) = \text{OPT}(0, i) = i\delta$ for all i, since the only way to line up an i-letter word with a 0-letter word is to use i gaps.

```
Alignment(X,Y)
  Array A[0...m, 0...n]
  Initialize A[i, 0] = iδ for each i
  Initialize A[0, j] = jδ for each j
  For j = 1, ..., n
     For i = 1, ..., m
         Use the recurrence (6.16) to compute A[i, j]
     Endfor
  Endfor
  Return A[m, n]
```

As in previous dynamic programming algorithms, we can trace back through the array A, using the second part of fact (6.16), to construct the alignment itself.

Analyzing the Algorithm

The correctness of the algorithm follows directly from (6.16). The running time is $O(mn)$, since the array A has $O(mn)$ entries, and at worst we spend constant time on each.

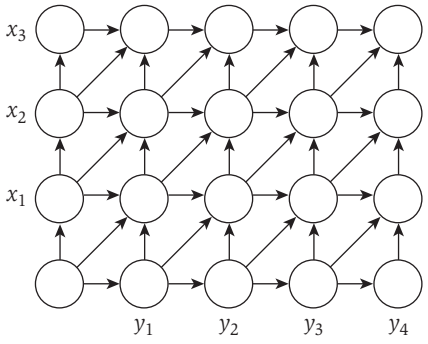


Figure 6.17 A graph-based picture of sequence alignment.

There is an appealing pictorial way in which people think about this sequence alignment algorithm. Suppose we build a two-dimensional $m \times n$ grid graph G_{XY}, with the rows labeled by symbols in the string X, the columns labeled by symbols in Y, and directed edges as in Figure 6.17.

We number the rows from 0 to m and the columns from 0 to n; we denote the node in the i^{th} row and the j^{th} column by the label (i, j). We put *costs* on the edges of G_{XY}: the cost of each horizontal and vertical edge is δ, and the cost of the diagonal edge from $(i-1, j-1)$ to (i, j) is $\alpha_{x_i y_j}$.

The purpose of this picture now emerges: the recurrence in (6.16) for $\text{OPT}(i, j)$ is precisely the recurrence one gets for the minimum-cost path in G_{XY} from $(0, 0)$ to (i, j). Thus we can show

(6.17) *Let $f(i, j)$ denote the minimum cost of a path from $(0, 0)$ to (i, j) in G_{XY}. Then for all i, j, we have $f(i, j) = \text{OPT}(i, j)$.*

Proof. We can easily prove this by induction on $i + j$. When $i + j = 0$, we have $i = j = 0$, and indeed $f(i, j) = \text{OPT}(i, j) = 0$.

Now consider arbitrary values of i and j, and suppose the statement is true for all pairs (i', j') with $i' + j' < i + j$. The last edge on the shortest path to (i, j) is either from $(i-1, j-1)$, $(i-1, j)$, or $(i, j-1)$. Thus we have

$$\begin{aligned} f(i, j) &= \min[\alpha_{x_i y_j} + f(i-1, j-1), \delta + f(i-1, j), \delta + f(i, j-1)] \\ &= \min[\alpha_{x_i y_j} + \text{OPT}(i-1, j-1), \delta + \text{OPT}(i-1, j), \delta + \text{OPT}(i, j-1)] \\ &= \text{OPT}(i, j), \end{aligned}$$

where we pass from the first line to the second using the induction hypothesis, and we pass from the second to the third using (6.16). ■

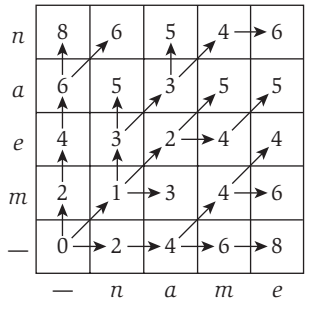


Figure 6.18 The OPT values for the problem of aligning the words *mean* to *name*.

Thus the value of the optimal alignment is the length of the shortest path in G_{XY} from $(0, 0)$ to (m, n). (We'll call any path in G_{XY} from $(0, 0)$ to (m, n) a *corner-to-corner path*.) Moreover, the diagonal edges used in a shortest path correspond precisely to the pairs used in a minimum-cost alignment. These connections to the Shortest-Path Problem in the graph G_{XY} do not directly yield an improvement in the running time for the sequence alignment problem; however, they do help one's intuition for the problem and have been useful in suggesting algorithms for more complex variations on sequence alignment.

For an example, Figure 6.18 shows the value of the shortest path from $(0, 0)$ to each node (i, j) for the problem of aligning the words *mean* and *name*. For the purpose of this example, we assume that $\delta = 2$; matching a vowel with a different vowel, or a consonant with a different consonant, costs 1; while matching a vowel and a consonant with each other costs 3. For each cell in the table (representing the corresponding node), the arrow indicates the last step of the shortest path leading to that node—in other words, the way that the minimum is achieved in (6.16). Thus, by following arrows backward from node $(4, 4)$, we can trace back to construct the alignment.

6.7 Sequence Alignment in Linear Space via Divide and Conquer

In the previous section, we showed how to compute the optimal alignment between two strings X and Y of lengths m and n, respectively. Building up the two-dimensional m-by-n array of optimal solutions to subproblems, OPT$(\cdot, \cdot)$, turned out to be equivalent to constructing a graph G_{XY} with mn nodes laid out in a grid and looking for the cheapest path between opposite corners. In either of these ways of formulating the dynamic programming algorithm, the running time is $O(mn)$, because it takes constant time to determine the value in each of the mn cells of the array OPT; and the space requirement is $O(mn)$ as well, since it was dominated by the cost of storing the array (or the graph G_{XY}).

The Problem

The question we ask in this section is: Should we be happy with $O(mn)$ as a space bound? If our application is to compare English words, or even English sentences, it is quite reasonable. In biological applications of sequence alignment, however, one often compares very long strings against one another; and in these cases, the $\Theta(mn)$ space requirement can potentially be a more severe problem than the $\Theta(mn)$ time requirement. Suppose, for example, that we are comparing two strings of 100,000 symbols each. Depending on the underlying processor, the prospect of performing roughly 10 billion primitive

operations might be less cause for worry than the prospect of working with a single 10-gigabyte array.

Fortunately, this is not the end of the story. In this section we describe a very clever enhancement of the sequence alignment algorithm that makes it work in $O(mn)$ time using only $O(m+n)$ space. In other words, we can bring the space requirement down to linear while blowing up the running time by at most an additional constant factor. For ease of presentation, we'll describe various steps in terms of paths in the graph G_{XY}, with the natural equivalence back to the sequence alignment problem. Thus, when we seek the pairs in an optimal alignment, we can equivalently ask for the edges in a shortest corner-to-corner path in G_{XY}.

The algorithm itself will be a nice application of divide-and-conquer ideas. The crux of the technique is the observation that, if we divide the problem into several recursive calls, then the space needed for the computation can be reused from one call to the next. The way in which this idea is used, however, is fairly subtle.

Designing the Algorithm

We first show that if we only care about the *value* of the optimal alignment, and not the alignment itself, it is easy to get away with linear space. The crucial observation is that to fill in an entry of the array A, the recurrence in (6.16) only needs information from the current column of A and the previous column of A. Thus we will "collapse" the array A to an $m \times 2$ array B: as the algorithm iterates through values of j, entries of the form $B[i,0]$ will hold the "previous" column's value $A[i,j-1]$, while entries of the form $B[i,1]$ will hold the "current" column's value $A[i,j]$.

```
Space-Efficient-Alignment(X,Y)
  Array B[0...m, 0...1]
  Initialize B[i,0] = iδ for each i (just as in column 0 of A)
  For j = 1, ..., n
     B[0,1] = jδ (since this corresponds to entry A[0,j])
     For i = 1, ..., m
          B[i,1] = min[α_{x_i y_j} + B[i-1,0],
                       δ + B[i-1,1],  δ + B[i,0]]
     Endfor
     Move column 1 of B to column 0 to make room for next iteration:
          Update B[i,0] = B[i,1] for each i
  Endfor
```

It is easy to verify that when this algorithm completes, the array entry $B[i, 1]$ holds the value of $\text{OPT}(i, n)$ for $i = 0, 1, \ldots, m$. Moreover, it uses $O(mn)$ time and $O(m)$ space. The problem is: where is the alignment itself? We haven't left enough information around to be able to run a procedure like `Find-Alignment`. Since B at the end of the algorithm only contains the last two columns of the original dynamic programming array A, if we were to try tracing back to get the path, we'd run out of information after just these two columns. We could imagine getting around this difficulty by trying to "predict" what the alignment is going to be in the process of running our space-efficient procedure. In particular, as we compute the values in the j^{th} column of the (now implicit) array A, we could try hypothesizing that a certain entry has a very small value, and hence that the alignment that passes through this entry is a promising candidate to be the optimal one. But this promising alignment might run into big problems later on, and a different alignment that currently looks much less attractive could turn out to be the optimal one.

There is, in fact, a solution to this problem—we will be able to recover the alignment itself using $O(m + n)$ space—but it requires a genuinely new idea. The insight is based on employing the divide-and-conquer technique that we've seen earlier in the book. We begin with a simple alternative way to implement the basic dynamic programming solution.

A Backward Formulation of the Dynamic Program Recall that we use $f(i, j)$ to denote the length of the shortest path from $(0, 0)$ to (i, j) in the graph G_{XY}. (As we showed in the initial sequence alignment algorithm, $f(i, j)$ has the same value as $\text{OPT}(i, j)$.) Now let's define $g(i, j)$ to be the length of the shortest path from (i, j) to (m, n) in G_{XY}. The function g provides an equally natural dynamic programming approach to sequence alignment, except that we build it up in reverse: we start with $g(m, n) = 0$, and the answer we want is $g(0, 0)$. By strict analogy with (6.16), we have the following recurrence for g.

(6.18) *For $i < m$ and $j < n$ we have*

$$g(i, j) = \min[\alpha_{x_{i+1}y_{j+1}} + g(i + 1, j + 1), \delta + g(i, j + 1), \delta + g(i + 1, j)].$$

This is just the recurrence one obtains by taking the graph G_{XY}, "rotating" it so that the node (m, n) is in the lower left corner, and using the previous approach. Using this picture, we can also work out the full dynamic programming algorithm to build up the values of g, *backward* starting from (m, n). Similarly, there is a space-efficient version of this backward dynamic programming algorithm, analogous to `Space-Efficient-Alignment`, which computes the value of the optimal alignment using only $O(m + n)$ space. We will refer to

this backward version, naturally enough, as `Backward-Space-Efficient-Alignment`.

Combining the Forward and Backward Formulations So now we have symmetric algorithms which build up the values of the functions f and g. The idea will be to use these two algorithms in concert to find the optimal alignment. First, here are two basic facts summarizing some relationships between the functions f and g.

(6.19) *The length of the shortest corner-to-corner path in G_{XY} that passes through (i, j) is $f(i, j) + g(i, j)$.*

Proof. Let ℓ_{ij} denote the length of the shortest corner-to-corner path in G_{XY} that passes through (i, j). Clearly, any such path must get from $(0, 0)$ to (i, j) and then from (i, j) to (m, n). Thus its length is at least $f(i, j) + g(i, j)$, and so we have $\ell_{ij} \geq f(i, j) + g(i, j)$. On the other hand, consider the corner-to-corner path that consists of a minimum-length path from $(0, 0)$ to (i, j), followed by a minimum-length path from (i, j) to (m, n). This path has length $f(i, j) + g(i, j)$, and so we have $\ell_{ij} \leq f(i, j) + g(i, j)$. It follows that $\ell_{ij} = f(i, j) + g(i, j)$. ■

(6.20) *Let k be any number in $\{0, \ldots, n\}$, and let q be an index that minimizes the quantity $f(q, k) + g(q, k)$. Then there is a corner-to-corner path of minimum length that passes through the node (q, k).*

Proof. Let ℓ^* denote the length of the shortest corner-to-corner path in G_{XY}. Now fix a value of $k \in \{0, \ldots, n\}$. The shortest corner-to-corner path must use *some* node in the k^{th} column of G_{XY}—let's suppose it is node (p, k)—and thus by (6.19)

$$\ell^* = f(p, k) + g(p, k) \geq \min_q f(q, k) + g(q, k).$$

Now consider the index q that achieves the minimum in the right-hand side of this expression; we have

$$\ell^* \geq f(q, k) + g(q, k).$$

By (6.19) again, the shortest corner-to-corner path using the node (q, k) has length $f(q, k) + g(q, k)$, and since ℓ^* is the minimum length of *any* corner-to-corner path, we have

$$\ell^* \leq f(q, k) + g(q, k).$$

It follows that $\ell^* = f(q, k) + g(q, k)$. Thus the shortest corner-to-corner path using the node (q, k) has length ℓ^*, and this proves (6.20). ■

Using (6.20) and our space-efficient algorithms to compute the *value* of the optimal alignment, we will proceed as follows. We divide G_{XY} along its center column and compute the value of $f(i, n/2)$ and $g(i, n/2)$ for each value of i, using our two space-efficient algorithms. We can then determine the minimum value of $f(i, n/2) + g(i, n/2)$, and conclude via (6.20) that there is a shortest corner-to-corner path passing through the node $(i, n/2)$. Given this, we can search for the shortest path recursively in the portion of G_{XY} between $(0, 0)$ and $(i, n/2)$ and in the portion between $(i, n/2)$ and (m, n). The crucial point is that we apply these recursive calls sequentially and reuse the working space from one call to the next. Thus, since we only work on one recursive call at a time, the total space usage is $O(m + n)$. The key question we have to resolve is whether the running time of this algorithm remains $O(mn)$.

In running the algorithm, we maintain a globally accessible list P which will hold nodes on the shortest corner-to-corner path as they are discovered. Initially, P is empty. P need only have $m + n$ entries, since no corner-to-corner path can use more than this many edges. We also use the following notation: $X[i : j]$, for $1 \le i \le j \le m$, denotes the substring of X consisting of $x_i x_{i+1} \cdots x_j$; and we define $Y[i : j]$ analogously. We will assume for simplicity that n is a power of 2; this assumption makes the discussion much cleaner, although it can be easily avoided.

```
Divide-and-Conquer-Alignment(X,Y)
  Let m be the number of symbols in X
  Let n be the number of symbols in Y
  If m ≤ 2 or n ≤ 2 then
     Compute optimal alignment using Alignment(X,Y)
  Call Space-Efficient-Alignment(X,Y[1 : n/2])
  Call Backward-Space-Efficient-Alignment(X,Y[n/2 + 1 : n])
  Let q be the index minimizing f(q, n/2) + g(q, n/2)
  Add (q, n/2) to global list P
  Divide-and-Conquer-Alignment(X[1 : q],Y[1 : n/2])
  Divide-and-Conquer-Alignment(X[q + 1 : n],Y[n/2 + 1 : n])
  Return P
```

As an example of the first level of recursion, consider Figure 6.19. If the *minimizing index* q turns out to be 1, we get the two subproblems pictured.

Analyzing the Algorithm

The previous arguments already establish that the algorithm returns the correct answer and that it uses $O(m + n)$ space. Thus, we need only verify the following fact.

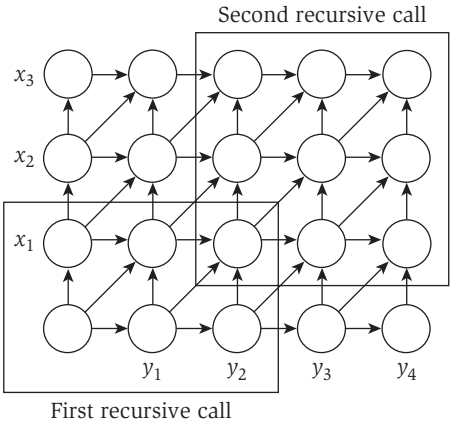


Figure 6.19 The first level of recurrence for the space-efficient `Divide-and-Conquer-Alignment`. The two boxed regions indicate the input to the two recursive cells.

(6.21) *The running time of* `Divide-and-Conquer-Alignment` *on strings of length m and n is $O(mn)$.*

Proof. Let $T(m, n)$ denote the maximum running time of the algorithm on strings of length m and n. The algorithm performs $O(mn)$ work to build up the arrays B and B'; it then runs recursively on strings of size q and $n/2$, and on strings of size $m - q$ and $n/2$. Thus, for some constant c, and some choice of index q, we have

$$T(m, n) \leq cmn + T(q, n/2) + T(m - q, n/2)$$
$$T(m, 2) \leq cm$$
$$T(2, n) \leq cn.$$

This recurrence is more complex than the ones we've seen in our earlier applications of divide-and-conquer in Chapter 5. First of all, the running time is a function of two variables (m and n) rather than just one; also, the division into subproblems is not necessarily an "even split," but instead depends on the value q that is found through the earlier work done by the algorithm.

So how should we go about solving such a recurrence? One way is to try guessing the form by considering a special case of the recurrence, and then using partial substitution to fill out the details of this guess. Specifically, suppose that we were in a case in which $m = n$, and in which the split point q were exactly in the middle. In this (admittedly restrictive) special case, we could write the function $T(\cdot)$ in terms of the single variable n, set $q = n/2$ (since we're assuming a perfect bisection), and have

$$T(n) \leq 2T(n/2) + cn^2.$$

This is a useful expression, since it's something that we solved in our earlier discussion of recurrences at the outset of Chapter 5. Specifically, this recurrence implies $T(n) = O(n^2)$.

So when $m = n$ and we get an even split, the running time grows like the square of n. Motivated by this, we move back to the fully general recurrence for the problem at hand and guess that $T(m, n)$ grows like the product of m and n. Specifically, we'll guess that $T(m, n) \leq kmn$ for some constant k, and see if we can prove this by induction. To start with the base cases $m \leq 2$ and $n \leq 2$, we see that these hold as long as $k \geq c/2$. Now, assuming $T(m', n') \leq km'n'$ holds for pairs (m', n') with a smaller product, we have

$$\begin{aligned} T(m, n) &\leq cmn + T(q, n/2) + T(m - q, n/2) \\ &\leq cmn + kqn/2 + k(m - q)n/2 \\ &= cmn + kqn/2 + kmn/2 - kqn/2 \\ &= (c + k/2)mn. \end{aligned}$$

Thus the inductive step will work if we choose $k = 2c$, and this completes the proof. ■

6.8 Shortest Paths in a Graph

For the final three sections, we focus on the problem of finding shortest paths in a graph, together with some closely related issues.

The Problem

Let $G = (V, E)$ be a directed graph. Assume that each edge $(i, j) \in E$ has an associated *weight* c_{ij}. The weights can be used to model a number of different things; we will picture here the interpretation in which the weight c_{ij} represents a *cost* for going directly from node i to node j in the graph.

Earlier we discussed Dijkstra's Algorithm for finding shortest paths in graphs with positive edge costs. Here we consider the more complex problem in which we seek shortest paths when costs may be negative. Among the motivations for studying this problem, here are two that particularly stand out. First, negative costs turn out to be crucial for modeling a number of phenomena with shortest paths. For example, the nodes may represent agents in a financial setting, and c_{ij} represents the cost of a transaction in which we buy from agent i and then immediately sell to agent j. In this case, a path would represent a succession of transactions, and edges with negative costs would represent transactions that result in profits. Second, the algorithm that we develop for dealing with edges of negative cost turns out, in certain crucial ways, to be more flexible and *decentralized* than Dijkstra's Algorithm. As a consequence, it has important applications for the design of distributed

routing algorithms that determine the most efficient path in a communication network.

In this section and the next two, we will consider the following two related problems.

- Given a graph G with weights, as described above, decide if G has a negative cycle—that is, a directed cycle C such that

$$\sum_{ij \in C} c_{ij} < 0.$$

- If the graph has no negative cycles, find a path P from an origin node s to a destination node t with minimum total cost:

$$\sum_{ij \in P} c_{ij}$$

  should be as small as possible for any s-t path. This is generally called both the *Minimum-Cost Path Problem* and the *Shortest-Path Problem*.

In terms of our financial motivation above, a negative cycle corresponds to a profitable sequence of transactions that takes us back to our starting point: we buy from i_1, sell to i_2, buy from i_2, sell to i_3, and so forth, finally arriving back at i_1 with a net profit. Thus negative cycles in such a network can be viewed as good *arbitrage opportunities*.

It makes sense to consider the minimum-cost s-t path problem under the assumption that there are no negative cycles. As illustrated by Figure 6.20, if there is a negative cycle C, a path P_s from s to the cycle, and another path P_t from the cycle to t, then we can build an s-t path of arbitrarily negative cost: we first use P_s to get to the negative cycle C, then we go around C as many times as we want, and then we use P_t to get from C to the destination t.

Designing and Analyzing the Algorithm

A Few False Starts Let's begin by recalling Dijkstra's Algorithm for the Shortest-Path Problem when there are no negative costs. That method

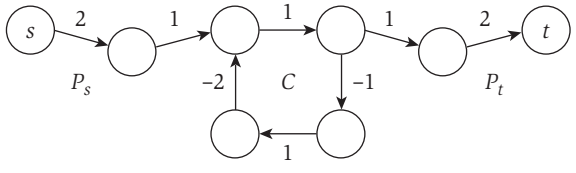
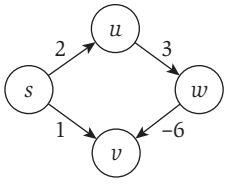
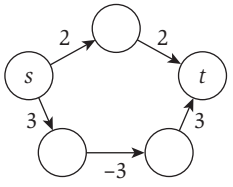


Figure 6.20 In this graph, one can find s-t paths of arbitrarily negative cost (by going around the cycle C many times).



(a)



(b)

Figure 6.21 (a) With negative edge costs, Dijkstra's Algorithm can give the wrong answer for the Shortest-Path Problem. (b) Adding 3 to the cost of each edge will make all edges nonnegative, but it will change the identity of the shortest s-t path.

computes a shortest path from the origin s to every other node v in the graph, essentially using a greedy algorithm. The basic idea is to maintain a set S with the property that the shortest path from s to each node in S is known. We start with $S = \{s\}$—since we know the shortest path from s to s has cost 0 when there are no negative edges—and we add elements greedily to this set S. As our first greedy step, we consider the minimum-cost edge leaving node s, that is, $\min_{i \in V} c_{si}$. Let v be a node on which this minimum is obtained. A key observation underlying Dijkstra's Algorithm is that the shortest path from s to v is the single-edge path $\{s, v\}$. Thus we can immediately add the node v to the set S. The path $\{s, v\}$ is clearly the shortest to v if there are no negative edge costs: any other path from s to v would have to start on an edge out of s that is at least as expensive as edge (s, v).

The above observation is no longer true if we can have negative edge costs. As suggested by the example in Figure 6.21(a), a path that starts on an expensive edge, but then compensates with subsequent edges of negative cost, can be cheaper than a path that starts on a cheap edge. This suggests that the Dijkstra-style greedy approach will not work here.

Another natural idea is to first modify the costs c_{ij} by adding some large constant M to each; that is, we let $c'_{ij} = c_{ij} + M$ for each edge $(i, j) \in E$. If the constant M is large enough, then all modified costs are nonnegative, and we can use Dijkstra's Algorithm to find the minimum-cost path subject to costs c'. However, this approach fails to find the correct minimum-cost paths with respect to the original costs c. The problem here is that changing the costs from c to c' changes the minimum-cost path. For example (as in Figure 6.21(b)), if a path P consisting of three edges is only slightly cheaper than another path P' that has two edges, then after the change in costs, P' will be cheaper, since we only add $2M$ to the cost of P' while adding $3M$ to the cost of P.

A Dynamic Programming Approach We will try to use dynamic programming to solve the problem of finding a shortest path from s to t when there are negative edge costs but no negative cycles. We could try an idea that has worked for us so far: subproblem i could be to find a shortest path using only the first i nodes. This idea does not immediately work, but it can be made to work with some effort. Here, however, we will discuss a simpler and more efficient solution, the *Bellman-Ford Algorithm*. The development of dynamic programming as a general algorithmic technique is often credited to the work of Bellman in the 1950's; and the Bellman-Ford Shortest-Path Algorithm was one of the first applications.

The dynamic programming solution we develop will be based on the following crucial observation.

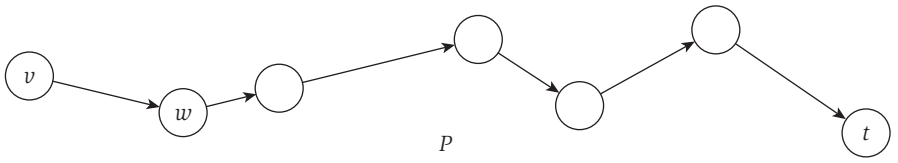


Figure 6.22 The minimum-cost path P from v to t using at most i edges.

(6.22) *If G has no negative cycles, then there is a shortest path from s to t that is simple (i.e., does not repeat nodes), and hence has at most $n - 1$ edges.*

Proof. Since every cycle has nonnegative cost, the shortest path P from s to t with the fewest number of edges does not repeat any vertex v. For if P did repeat a vertex v, we could remove the portion of P between consecutive visits to v, resulting in a path of no greater cost and fewer edges. ■

Let's use $\text{OPT}(i, v)$ to denote the minimum cost of a v-t path using at most i edges. By (6.22), our original problem is to compute $\text{OPT}(n-1, s)$. (We could instead design an algorithm whose subproblems correspond to the minimum cost of an s-v path using at most i edges. This would form a more natural parallel with Dijkstra's Algorithm, but it would not be as natural in the context of the routing protocols we discuss later.)

We now need a simple way to express $\text{OPT}(i, v)$ using smaller subproblems. We will see that the most natural approach involves the consideration of many different options; this is another example of the principle of "multi-way choices" that we saw in the algorithm for the Segmented Least Squares Problem.

Let's fix an optimal path P representing $\text{OPT}(i, v)$ as depicted in Figure 6.22.

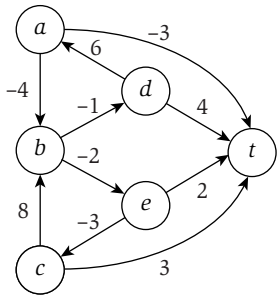
- If the path P uses at most $i - 1$ edges, then $\text{OPT}(i, v) = \text{OPT}(i - 1, v)$.
- If the path P uses i edges, and the first edge is (v, w), then $\text{OPT}(i, v) = c_{vw} + \text{OPT}(i - 1, w)$.

This leads to the following recursive formula.

(6.23) *If $i > 0$ then*

$$\text{OPT}(i, v) = \min(\text{OPT}(i-1, v), \min_{w \in V}(\text{OPT}(i-1, w) + c_{vw})).$$

Using this recurrence, we get the following dynamic programming algorithm to compute the value $\text{OPT}(n-1, s)$.



(a)

	0	1	2	3	4	5
t	0	0	0	0	0	0
a	∞	−3	−3	−4	−6	−6
b	∞	∞	0	−2	−2	−2
c	∞	3	3	3	3	3
d	∞	4	3	3	2	0
e	∞	2	0	0	0	0

(b)

Figure 6.23 For the directed graph in (a), the Shortest-Path Algorithm constructs the dynamic programming table in (b).

```
Shortest-Path(G, s, t)
  n = number of nodes in G
  Array M[0 ... n − 1, V]
  Define M[0, t] = 0 and M[0, v] = ∞ for all other v ∈ V
  For i = 1, ..., n − 1
    For v ∈ V in any order
      Compute M[i, v] using the recurrence (6.23)
    Endfor
  Endfor
  Return M[n − 1, s]
```

The correctness of the method follows directly by induction from (6.23). We can bound the running time as follows. The table M has n^2 entries; and each entry can take $O(n)$ time to compute, as there are at most n nodes $w \in V$ we have to consider.

(6.24) *The* `Shortest-Path` *method correctly computes the minimum cost of an s-t path in any graph that has no negative cycles, and runs in $O(n^3)$ time.*

Given the table M containing the optimal values of the subproblems, the shortest path using at most i edges can be obtained in $O(in)$ time, by tracing back through smaller subproblems.

As an example, consider the graph in Figure 6.23(a), where the goal is to find a shortest path from each node to t. The table in Figure 6.23(b) shows the array M, with entries corresponding to the values $M[i, v]$ from the algorithm. Thus a single row in the table corresponds to the shortest path from a particular node to t, as we allow the path to use an increasing number of edges. For example, the shortest path from node d to t is updated four times, as it changes from d-t, to d-a-t, to d-a-b-e-t, and finally to d-a-b-e-c-t.

Extensions: Some Basic Improvements to the Algorithm

An Improved Running-Time Analysis We can actually provide a better running-time analysis for the case in which the graph G does not have too many edges. A directed graph with n nodes can have close to n^2 edges, since there could potentially be an edge between each pair of nodes, but many graphs are much sparser than this. When we work with a graph for which the number of edges m is significantly less than n^2, we've already seen in a number of cases earlier in the book that it can be useful to write the running-time in terms of both m and n; this way, we can quantify our speed-up on graphs with relatively fewer edges.

If we are a little more careful in the analysis of the method above, we can improve the running-time bound to $O(mn)$ without significantly changing the algorithm itself.

(6.25) *The* `Shortest-Path` *method can be implemented in* $O(mn)$ *time.*

Proof. Consider the computation of the array entry $M[i, v]$ according to the recurrence (6.23); we have

$$M[i, v] = \min(M[i-1, v], \min_{w \in V}(M[i-1, w] + c_{vw})).$$

We assumed it could take up to $O(n)$ time to compute this minimum, since there are n possible nodes w. But, of course, we need only compute this minimum over all nodes w for which v has an edge to w; let us use n_v to denote this number. Then it takes time $O(n_v)$ to compute the array entry $M[i, v]$. We have to compute an entry for every node v and every index $0 \leq i \leq n-1$, so this gives a running-time bound of

$$O\left(n \sum_{v \in V} n_v\right).$$

In Chapter 3, we performed exactly this kind of analysis for other graph algorithms, and used (3.9) from that chapter to bound the expression $\sum_{v \in V} n_v$ for undirected graphs. Here we are dealing with directed graphs, and n_v denotes the number of edges *leaving* v. In a sense, it is even easier to work out the value of $\sum_{v \in V} n_v$ for the directed case: each edge leaves exactly one of the nodes in V, and so each edge is counted exactly once by this expression. Thus we have $\sum_{v \in V} n_v = m$. Plugging this into our expression

$$O\left(n \sum_{v \in V} n_v\right)$$

for the running time, we get a running-time bound of $O(mn)$. ■

Improving the Memory Requirements We can also significantly improve the memory requirements with only a small change to the implementation. A common problem with many dynamic programming algorithms is the large space usage, arising from the M array that needs to be stored. In the Bellman-Ford Algorithm as written, this array has size n^2; however, we now show how to reduce this to $O(n)$. Rather than recording $M[i, v]$ for each value i, we will use and update a single value $M[v]$ for each node v, the length of the shortest path from v to t that we have found so far. We still run the algorithm for

iterations $i = 1, 2, \ldots, n-1$, but the role of i will now simply be as a counter; in each iteration, and for each node v, we perform the update

$$M[v] = \min(M[v], \min_{w \in V}(c_{vw} + M[w])).$$

We now observe the following fact.

(6.26) *Throughout the algorithm $M[v]$ is the length of some path from v to t, and after i rounds of updates the value $M[v]$ is no larger than the length of the shortest path from v to t using at most i edges.*

Given (6.26), we can then use (6.22) as before to show that we are done after $n-1$ iterations. Since we are only storing an M array that indexes over the nodes, this requires only $O(n)$ working memory.

Finding the Shortest Paths One issue to be concerned about is whether this space-efficient version of the algorithm saves enough information to recover the shortest paths themselves. In the case of the Sequence Alignment Problem in the previous section, we had to resort to a tricky divide-and-conquer method to recover the solution from a similar space-efficient implementation. Here, however, we will be able to recover the shortest paths much more easily.

To help with recovering the shortest paths, we will enhance the code by having each node v maintain the first node (after itself) on its path to the destination t; we will denote this first node by $first[v]$. To maintain $first[v]$, we update its value whenever the distance $M[v]$ is updated. In other words, whenever the value of $M[v]$ is reset to the minimum $\min_{w \in V}(c_{vw} + M[w])$, we set $first[v]$ to the node w that attains this minimum.

Now let P denote the directed "pointer graph" whose nodes are V, and whose edges are $\{(v, first[v])\}$. The main observation is the following.

(6.27) *If the pointer graph P contains a cycle C, then this cycle must have negative cost.*

Proof. Notice that if $first[v] = w$ at any time, then we must have $M[v] \geq c_{vw} + M[w]$. Indeed, the left- and right-hand sides are equal after the update that sets $first[v]$ equal to w; and since $M[w]$ may decrease, this equation may turn into an inequality.

Let $v_1, v_2, \ldots, v_k$ be the nodes along the cycle C in the pointer graph, and assume that (v_k, v_1) is the last edge to have been added. Now, consider the values right before this last update. At this time we have $M[v_i] \geq c_{v_i v_{i+1}} + M[v_{i+1}]$ for all $i = 1, \ldots, k-1$, and we also have $M[v_k] > c_{v_k v_1} + M[v_1]$ since we are about to update $M[v_k]$ and change $first[v_k]$ to v_1. Adding all these inequalities, the $M[v_i]$ values cancel, and we get $0 > \sum_{i=1}^{k-1} c_{v_i v_{i+1}} + c_{v_k v_1}$: a negative cycle, as claimed. ■

Now note that if G has no negative cycles, then (6.27) implies that the pointer graph P will never have a cycle. For a node v, consider the path we get by following the edges in P, from v to $first[v] = v_1$, to $first[v_1] = v_2$, and so forth. Since the pointer graph has no cycles, and the sink t is the only node that has no outgoing edge, this path must lead to t. We claim that when the algorithm terminates, this is in fact a shortest path in G from v to t.

(6.28) *Suppose G has no negative cycles, and consider the pointer graph P at the termination of the algorithm. For each node v, the path in P from v to t is a shortest v-t path in G.*

Proof. Consider a node v and let $w = first[v]$. Since the algorithm terminated, we must have $M[v] = c_{vw} + M[w]$. The value $M[t] = 0$, and hence the length of the path traced out by the pointer graph is exactly $M[v]$, which we know is the shortest-path distance. ■

Note that in the more space-efficient version of Bellman-Ford, the path whose length is $M[v]$ after i iterations can have substantially more edges than i. For example, if the graph is a single path from s to t, and we perform updates in the reverse of the order the edges appear on the path, then we get the final shortest-path values in just one iteration. This does not always happen, so we cannot claim a worst-case running-time improvement, but it would be nice to be able to use this fact opportunistically to speed up the algorithm on instances where it does happen. In order to do this, we need a stopping signal in the algorithm—something that tells us it's safe to terminate before iteration $n - 1$ is reached.

Such a stopping signal is a simple consequence of the following observation: If we ever execute a complete iteration i in which *no* $M[v]$ value changes, then no $M[v]$ value will ever change again, since future iterations will begin with exactly the same set of array entries. Thus it is safe to stop the algorithm. Note that it is not enough for a *particular* $M[v]$ value to remain the same; in order to safely terminate, we need for all these values to remain the same for a single iteration.

6.9 Shortest Paths and Distance Vector Protocols

One important application of the Shortest-Path Problem is for routers in a communication network to determine the most efficient path to a destination. We represent the network using a graph in which the nodes correspond to routers, and there is an edge between v and w if the two routers are connected by a direct communication link. We define a cost c_{vw} representing the delay on the link (v, w); the Shortest-Path Problem with these costs is to determine the path with minimum delay from a source node s to a destination t. Delays are

naturally nonnegative, so one could use Dijkstra's Algorithm to compute the shortest path. However, Dijkstra's shortest-path computation requires global knowledge of the network: it needs to maintain a set S of nodes for which shortest paths have been determined, and make a global decision about which node to add next to S. While routers can be made to run a protocol in the background that gathers enough global information to implement such an algorithm, it is often cleaner and more flexible to use algorithms that require only local knowledge of neighboring nodes.

If we think about it, the Bellman-Ford Algorithm discussed in the previous section has just such a "local" property. Suppose we let each node v maintain its value $M[v]$; then to update this value, v needs only obtain the value $M[w]$ from each neighbor w, and compute

$$\min_{w \in V}(c_{vw} + M[w])$$

based on the information obtained.

We now discuss an improvement to the Bellman-Ford Algorithm that makes it better suited for routers and, at the same time, a faster algorithm in practice. Our current implementation of the Bellman-Ford Algorithm can be thought of as a *pull-based* algorithm. In each iteration i, each node v has to contact each neighbor w, and "pull" the new value $M[w]$ from it. If a node w has not changed its value, then there is no need for v to get the value again; however, v has no way of knowing this fact, and so it must execute the pull anyway.

This wastefulness suggests a symmetric *push-based* implementation, where values are only transmitted when they change. Specifically, each node w whose distance value $M[w]$ changes in an iteration informs all its neighbors of the new value in the next iteration; this allows them to update their values accordingly. If $M[w]$ has not changed, then the neighbors of w already have the current value, and there is no need to "push" it to them again. This leads to savings in the running time, as not all values need to be pushed in each iteration. We also may terminate the algorithm early, if no value changes during an iteration. Here is a concrete description of the push-based implementation.

```
Push-Based-Shortest-Path(G, s, t)
  n = number of nodes in G
  Array M[V]
  Initialize M[t] = 0 and M[v] = ∞ for all other v ∈ V
  For i = 1, ..., n − 1
    For w ∈ V in any order
      If M[w] has been updated in the previous iteration then
```

```
      For all edges (v, w) in any order
        M[v] = min(M[v], c_vw + M[w])
        If this changes the value of M[v], then first[v] = w
      Endfor
  Endfor
  If no value changed in this iteration, then end the algorithm
Endfor
Return M[s]
```

In this algorithm, nodes are sent updates of their neighbors' distance values in rounds, and each node sends out an update in each iteration in which it has changed. However, if the nodes correspond to routers in a network, then we do not expect everything to run in lockstep like this; some routers may report updates much more quickly than others, and a router with an update to report may sometimes experience a delay before contacting its neighbors. Thus the routers will end up executing an *asynchronous* version of the algorithm: each time a node w experiences an update to its $M[w]$ value, it becomes "active" and eventually notifies its neighbors of the new value. If we were to watch the behavior of all routers interleaved, it would look as follows.

```
Asynchronous-Shortest-Path(G, s, t)
  n = number of nodes in G
  Array M[V]
  Initialize M[t] = 0 and M[v] = ∞ for all other v ∈ V
  Declare t to be active and all other nodes inactive
  While there exists an active node
    Choose an active node w
      For all edges (v, w) in any order
        M[v] = min(M[v], c_vw + M[w])
        If this changes the value of M[v], then
          first[v] = w
          v becomes active
      Endfor
      w becomes inactive
  EndWhile
```

One can show that even this version of the algorithm, with essentially no coordination in the ordering of updates, will converge to the correct values of the shortest-path distances to t, assuming only that each time a node becomes active, it eventually contacts its neighbors.

The algorithm we have developed here uses a single destination t, and all nodes $v \in V$ compute their shortest path to t. More generally, we are

presumably interested in finding distances and shortest paths between all pairs of nodes in a graph. To obtain such distances, we effectively use n separate computations, one for each destination. Such an algorithm is referred to as a *distance vector protocol*, since each node maintains a vector of distances to every other node in the network.

Problems with the Distance Vector Protocol

One of the major problems with the distributed implementation of Bellman-Ford on routers (the protocol we have been discussing above) is that it's derived from an initial dynamic programming algorithm that assumes edge costs will remain constant during the execution of the algorithm. Thus far we've been designing algorithms with the tacit understanding that a program executing the algorithm will be running on a single computer (or a centrally managed set of computers), processing some specified input. In this context, it's a rather benign assumption to require that the input not change while the program is actually running. Once we start thinking about routers in a network, however, this assumption becomes troublesome. Edge costs may change for all sorts of reasons: links can become congested and experience slow-downs; or a link (v, w) may even fail, in which case the cost c_{vw} effectively increases to ∞.

Here's an indication of what can go wrong with our shortest-path algorithm when this happens. If an edge (v, w) is deleted (say the link goes down), it is natural for node v to react as follows: it should check whether its shortest path to some node t used the edge (v, w), and, if so, it should increase the distance using other neighbors. Notice that this increase in distance from v can now trigger increases at v's neighbors, if they were relying on a path through v, and these changes can cascade through the network. Consider the extremely simple example in Figure 6.24, in which the original graph has three edges (s, v), (v, s) and (v, t), each of cost 1.

Now suppose the edge (v, t) in Figure 6.24 is deleted. How does node v react? Unfortunately, it does not have a global map of the network; it only knows the shortest-path distances of each of its neighbors to t. Thus it does

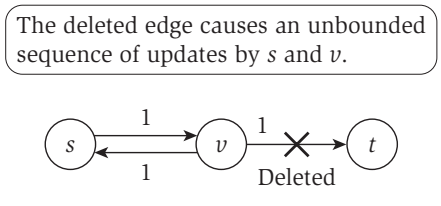


Figure 6.24 When the edge (v, t) is deleted, the distributed Bellman-Ford Algorithm will begin "counting to infinity."

not know that the deletion of (v, t) has eliminated all paths from s to t. Instead, it sees that $M[s] = 2$, and so it updates $M[v] = c_{vs} + M[s] = 3$, assuming that it will use its cost-1 edge to s, followed by the supposed cost-2 path from s to t. Seeing this change, node s will update $M[s] = c_{sv} + M[v] = 4$, based on its cost-1 edge to v, followed by the supposed cost-3 path from v to t. Nodes s and v will continue updating their distance to t until one of them finds an alternate route; in the case, as here, that the network is truly disconnected, these updates will continue indefinitely—a behavior known as the problem of *counting to infinity*.

To avoid this problem and related difficulties arising from the limited amount of information available to nodes in the Bellman-Ford Algorithm, the designers of network routing schemes have tended to move from distance vector protocols to more expressive *path vector protocols*, in which each node stores not just the distance and first hop of their path to a destination, but some representation of the entire path. Given knowledge of the paths, nodes can avoid updating their paths to use edges they know to be deleted; at the same time, they require significantly more storage to keep track of the full paths. In the history of the Internet, there has been a shift from distance vector protocols to path vector protocols; currently, the path vector approach is used in the *Border Gateway Protocol* (BGP) in the Internet core.

* 6.10 Negative Cycles in a Graph

So far in our consideration of the Bellman-Ford Algorithm, we have assumed that the underlying graph has negative edge costs but no negative cycles. We now consider the more general case of a graph that may contain negative cycles.

The Problem

There are two natural questions we will consider.

- How do we decide if a graph contains a negative cycle?
- How do we actually find a negative cycle in a graph that contains one?

The algorithm developed for finding negative cycles will also lead to an improved practical implementation of the Bellman-Ford Algorithm from the previous sections.

It turns out that the ideas we've seen so far will allow us to find negative cycles that have a path reaching a sink t. Before we develop the details of this, let's compare the problem of finding a negative cycle that can reach a given t with the seemingly more natural problem of finding a negative cycle *anywhere* in the graph, regardless of its position related to a sink. It turns out that if we

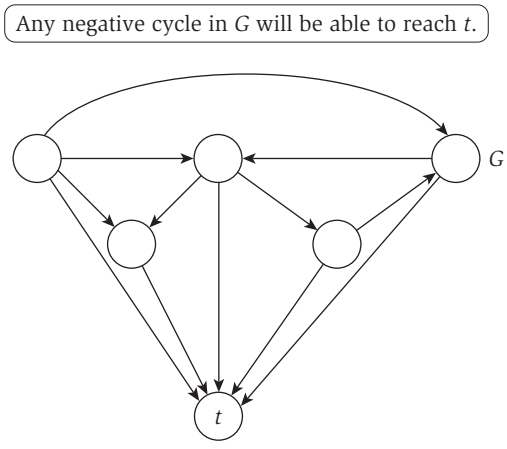


Figure 6.25 The augmented graph.

develop a solution to the first problem, we'll be able to obtain a solution to the second problem as well, in the following way. Suppose we start with a graph G, add a new node t to it, and connect each other node v in the graph to node t via an edge of cost 0, as shown in Figure 6.25. Let us call the new "augmented graph" G'.

(6.29) *The augmented graph G' has a negative cycle C such that there is a path from C to the sink t if and only if the original graph has a negative cycle.*

Proof. Assume G has a negative cycle. Then this cycle C clearly has an edge to t in G', since all nodes have an edge to t.

Now suppose G' has a negative cycle with a path to t. Since no edge leaves t in G', this cycle cannot contain t. Since G' is the same as G aside from the node t, it follows that this cycle is also a negative cycle of G. ■

So it is really enough to solve the problem of deciding whether G has a negative cycle that has a path to a given sink node t, and we do this now.

Designing and Analyzing the Algorithm

To get started thinking about the algorithm, we begin by adopting the original version of the Bellman-Ford Algorithm, which was less efficient in its use of space. We first extend the definitions of $\text{OPT}(i, v)$ from the Bellman-Ford Algorithm, defining them for values $i \geq n$. With the presence of a negative cycle in the graph, (6.22) no longer applies, and indeed the shortest path may

get shorter and shorter as we go around a negative cycle. In fact, for any node v on a negative cycle that has a path to t, we have the following.

(6.30) *If node v can reach node t and is contained in a negative cycle, then*

$$\lim_{i \to \infty} \text{OPT}(i, v) = -\infty.$$

If the graph has no negative cycles, then (6.22) implies following statement.

(6.31) *If there are no negative cycles in G, then $\text{OPT}(i, v) = \text{OPT}(n-1, v)$ for all nodes v and all $i \geq n$.*

But for how large an i do we have to compute the values $\text{OPT}(i, v)$ before concluding that the graph has no negative cycles? For example, a node v may satisfy the equation $\text{OPT}(n, v) = \text{OPT}(n-1, v)$, and yet still lie on a negative cycle. (Do you see why?) However, it turns out that we will be in good shape if this equation holds for all nodes.

(6.32) *There is no negative cycle with a path to t if and only if $\text{OPT}(n, v) = \text{OPT}(n-1, v)$ for all nodes v.*

Proof. Statement (6.31) has already proved the forward direction. For the other direction, we use an argument employed earlier for reasoning about when it's safe to stop the Bellman-Ford Algorithm early. Specifically, suppose $\text{OPT}(n, v) = \text{OPT}(n-1, v)$ for all nodes v. The values of $\text{OPT}(n+1, v)$ can be computed from $\text{OPT}(n, v)$; but all these values are the same as the corresponding $\text{OPT}(n-1, v)$. It follows that we will have $\text{OPT}(n+1, v) = \text{OPT}(n-1, v)$. Extending this reasoning to future iterations, we see that none of the values will ever change again, that is, $\text{OPT}(i, v) = \text{OPT}(n-1, v)$ for all nodes v and all $i \geq n$. Thus there cannot be a negative cycle C that has a path to t; for any node w on this cycle C, (6.30) implies that the values $\text{OPT}(i, w)$ would have to become arbitrarily negative as i increased. ■

Statement (6.32) gives an $O(mn)$ method to decide if G has a negative cycle that can reach t. We compute values of $\text{OPT}(i, v)$ for nodes of G and for values of i up to n. By (6.32), there is no negative cycle if and only if there is some value of $i \leq n$ at which $\text{OPT}(i, v) = \text{OPT}(i-1, v)$ for all nodes v.

So far we have determined whether or not the graph has a negative cycle with a path from the cycle to t, but we have not actually found the cycle. To find a negative cycle, we consider a node v such that $\text{OPT}(n, v) \neq \text{OPT}(n-1, v)$: for this node, a path P from v to t of cost $\text{OPT}(n, v)$ must use *exactly* n edges. We find this minimum-cost path P from v to t by tracing back through the subproblems. As in our proof of (6.22), a simple path can only have $n-1$

edges, so P must contain a cycle C. We claim that this cycle C has negative cost.

(6.33) *If G has n nodes and $\text{OPT}(n, v) \neq \text{OPT}(n-1, v)$, then a path P from v to t of cost $\text{OPT}(n, v)$ contains a cycle C, and C has negative cost.*

Proof. First observe that the path P must have n edges, as $\text{OPT}(n, v) \neq \text{OPT}(n-1, v)$, and so every path using $n-1$ edges has cost greater than that of the path P. In a graph with n nodes, a path consisting of n edges must repeat a node somewhere; let w be a node that occurs on P more than once. Let C be the cycle on P between two consecutive occurrences of node w. If C were not a negative cycle, then deleting C from P would give us a v-t path with fewer than n edges and no greater cost. This contradicts our assumption that $\text{OPT}(n, v) \neq \text{OPT}(n-1, v)$, and hence C must be a negative cycle. ■

(6.34) *The algorithm above finds a negative cycle in G, if such a cycle exists, and runs in $O(mn)$ time.*

Extensions: Improved Shortest Paths and Negative Cycle Detection Algorithms

At the end of Section 6.8 we discussed a space-efficient implementation of the Bellman-Ford algorithm for graphs with no negative cycles. Here we implement the detection of negative cycles in a comparably space-efficient way. In addition to the savings in space, this will also lead to a considerable speedup in practice even for graphs with no negative cycles. The implementation will be based on the same pointer graph P derived from the "first edges" $(v, \mathit{first}[v])$ that we used for the space-efficient implementation in Section 6.8. By (6.27), we know that if the pointer graph ever has a cycle, then the cycle has negative cost, and we are done. But if G has a negative cycle, does this guarantee that the pointer graph will ever have a cycle? Furthermore, how much extra computation time do we need for periodically checking whether P has a cycle?

Ideally, we would like to determine whether a cycle is created in the pointer graph P every time we add a new edge (v, w) with $\mathit{first}[v] = w$. An additional advantage of such "instant" cycle detection will be that we will not have to wait for n iterations to see that the graph has a negative cycle: We can terminate as soon as a negative cycle is found. Earlier we saw that if a graph G has no negative cycles, the algorithm can be stopped early if in some iteration the shortest path values $M[v]$ remain the same *for all nodes* v. Instant negative cycle detection will be an analogous early termination rule for graphs that have negative cycles.

Consider a new edge (v, w), with $first[v] = w$, that is added to the pointer graph P. Before we add (v, w) the pointer graph has no cycles, so it consists of paths from each node v to the sink t. The most natural way to check whether adding edge (v, w) creates a cycle in P is to follow the current path from w to the terminal t in time proportional to the length of this path. If we encounter v along this path, then a cycle has been formed, and hence, by (6.27), the graph has a negative cycle. Consider Figure 6.26, for example, where in both (a) and (b) the pointer $first[v]$ is being updated from u to w; in (a), this does not result in a (negative) cycle, but in (b) it does. However, if we trace out the sequence of pointers from v like this, then we could spend as much as $O(n)$ time following the path to t and still not find a cycle. We now discuss a method that does not require an $O(n)$ blow-up in the running time.

We know that before the new edge (v, w) was added, the pointer graph was a directed tree. Another way to test whether the addition of (v, w) creates a cycle is to consider all nodes in the subtree directed toward v. If w is in this subtree, then (v, w) forms a cycle; otherwise it does not. (Again, consider the two sample cases in Figure 6.26.) To be able to find all nodes in the subtree directed toward v, we need to have each node v maintain a list of all other nodes whose selected edges point to v. Given these pointers, we can find the subtree in time proportional to the size of the subtree pointing to v, at most $O(n)$ as before. However, here we will be able to make additional use of the work done. Notice that the current distance value $M[x]$ for all nodes x in the subtree was derived from node v's old value. We have just updated v's distance, and hence we know that the distance values of all these nodes will be updated again. We'll mark each of these nodes x as "dormant," delete the

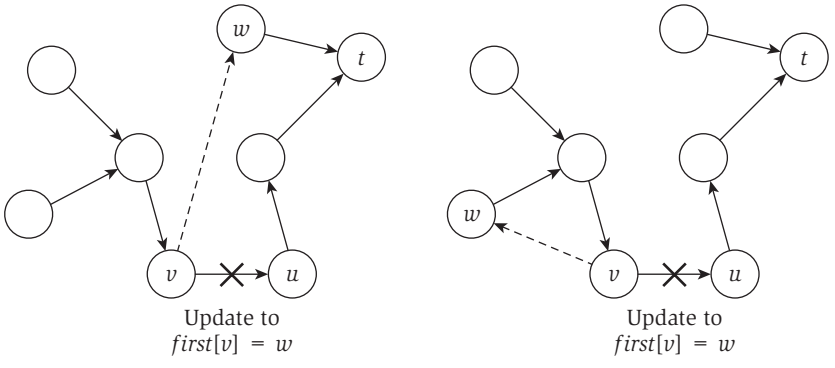


Figure 6.26 Changing the pointer graph P when $first[v]$ is updated from u to w. In (b), this creates a (negative) cycle, whereas in (a) it does not.

edge $(x, first[x])$ from the pointer graph, and not use x for future updates until its distance value changes.

This can save a lot of future work in updates, but what is the effect on the worst-case running time? We can spend as much as $O(n)$ extra time marking nodes dormant after every update in distances. However, a node can be marked dormant only if a pointer had been defined for it at some point in the past, so the time spent on marking nodes dormant is at most as much as the time the algorithm spends updating distances.

Now consider the time the algorithm spends on operations other than marking nodes dormant. Recall that the algorithm is divided into iterations, where iteration $i+1$ processes nodes whose distance has been updated in iteration i. For the original version of the algorithm, we showed in (6.26) that after i iterations, the value $M[v]$ is no larger than the value of the shortest path from v to t using at most i edges. However, with many nodes dormant in each iteration, this may not be true anymore. For example, if the shortest path from v to t using at most i edges starts on edge $e = (v, w)$, and w is dormant in this iteration, then we may not update the distance value $M[v]$, and so it stays at a value higher than the length of the path through the edge (v, w). This seems like a problem—however, in this case, the path through edge (v, w) is not actually the shortest path, so $M[v]$ will have a chance to get updated later to an even smaller value.

So instead of the simpler property that held for $M[v]$ in the original versions of the algorithm, we now have the the following claim.

(6.35) *Throughout the algorithm $M[v]$ is the length of some simple path from v to t; the path has at least i edges if the distance value $M[v]$ is updated in iteration i; and after i iterations, the value $M[v]$ is the length of the shortest path for all nodes v where there is a shortest v-t path using at most i edges.*

Proof. The *first* pointers maintain a tree of paths to t, which implies that all paths used to update the distance values are simple. The fact that updates in iteration i are caused by paths with at least i edges is easy to show by induction on i. Similarly, we use induction to show that after iteration i the value $M[v]$ is the distance on all nodes v where the shortest path from v to t uses at most i edges. Note that nodes v where $M[v]$ is the actual shortest-path distance cannot be dormant, as the value $M[v]$ will be updated in the next iteration for all dormant nodes. ■

Using this claim, we can see that the worst-case running time of the algorithm is still bounded by $O(mn)$: Ignoring the time spent on marking nodes dormant, each iteration is implemented in $O(m)$ time, and there can be at most $n-1$ iterations that update values in the array M without finding

a negative cycle, as simple paths can have at most $n - 1$ edges. Finally, the time spent marking nodes dormant is bounded by the time spent on updates. We summarize the discussion with the following claim about the worst-case performance of the algorithm. In fact, as mentioned above, this new version is in practice the fastest implementation of the algorithm even for graphs that do not have negative cycles, or even negative-cost edges.

(6.36) *The improved algorithm outlined above finds a negative cycle in G if such a cycle exists. It terminates immediately if the pointer graph P of first$[v]$ pointers contains a cycle C, or if there is an iteration in which no update occurs to any distance value $M[v]$. The algorithm uses $O(n)$ space, has at most n iterations, and runs in $O(mn)$ time in the worst case.*

Solved Exercises

Solved Exercise 1

Suppose you are managing the construction of billboards on the Stephen Daedalus Memorial Highway, a heavily traveled stretch of road that runs west-east for M miles. The possible sites for billboards are given by numbers $x_1, x_2, \ldots, x_n$, each in the interval $[0, M]$ (specifying their position along the highway, measured in miles from its western end). If you place a billboard at location x_i, you receive a revenue of $r_i > 0$.

Regulations imposed by the county's Highway Department require that no two of the billboards be within less than or equal to 5 miles of each other. You'd like to place billboards at a subset of the sites so as to maximize your total revenue, subject to this restriction.

Example. Suppose $M = 20$, $n = 4$,

$$\{x_1, x_2, x_3, x_4\} = \{6, 7, 12, 14\},$$

and

$$\{r_1, r_2, r_3, r_4\} = \{5, 6, 5, 1\}.$$

Then the optimal solution would be to place billboards at x_1 and x_3, for a total revenue of 10.

Give an algorithm that takes an instance of this problem as input and returns the maximum total revenue that can be obtained from any valid subset of sites. The running time of the algorithm should be polynomial in n.

Solution We can naturally apply dynamic programming to this problem if we reason as follows. Consider an optimal solution for a given input instance; in this solution, we either place a billboard at site x_n or not. If we don't, the optimal solution on sites $x_1, \ldots, x_n$ is really the same as the optimal solution

on sites $x_1, \ldots, x_{n-1}$; if we do, then we should eliminate x_n and all other sites that are within 5 miles of it, and find an optimal solution on what's left. The same reasoning applies when we're looking at the problem defined by just the first j sites, $x_1, \ldots, x_j$: we either include x_j in the optimal solution or we don't, with the same consequences.

Let's define some notation to help express this. For a site x_j, we let $e(j)$ denote the easternmost site x_i that is more than 5 miles from x_j. Since sites are numbered west to east, this means that the sites $x_1, x_2, \ldots, x_{e(j)}$ are still valid options once we've chosen to place a billboard at x_j, but the sites $x_{e(j)+1}, \ldots, x_{j-1}$ are not.

Now, our reasoning above justifies the following recurrence. If we let $\text{OPT}(j)$ denote the revenue from the optimal subset of sites among $x_1, \ldots, x_j$, then we have

$$\text{OPT}(j) = \max(r_j + \text{OPT}(e(j)), \text{OPT}(j-1)).$$

We now have most of the ingredients we need for a dynamic programming algorithm. First, we have a set of n subproblems, consisting of the first j sites for $j = 0, 1, 2, \ldots, n$. Second, we have a recurrence that lets us build up the solutions to subproblems, given by $\text{OPT}(j) = \max(r_j + \text{OPT}(e(j)), \text{OPT}(j-1))$.

To turn this into an algorithm, we just need to define an array M that will store the OPT values and throw a loop around the recurrence that builds up the values $M[j]$ in order of increasing j.

```
Initialize M[0] = 0 and M[1] = r_1
For j = 2, 3, ..., n:
  Compute M[j] using the recurrence
Endfor
Return M[n]
```

As with all the dynamic programming algorithms we've seen in this chapter, an optimal *set* of billboards can be found by tracing back through the values in array M.

Given the values $e(j)$ for all j, the running time of the algorithm is $O(n)$, since each iteration of the loop takes constant time. We can also compute all $e(j)$ values in $O(n)$ time as follows. For each site location x_i, we define $x'_i = x_i - 5$. We then merge the sorted list $x_1, \ldots, x_n$ with the sorted list $x'_1, \ldots, x'_n$ in linear time, as we saw how to do in Chapter 2. We now scan through this merged list; when we get to the entry x'_j, we know that anything from this point onward to x_j cannot be chosen together with x_j (since it's within 5 miles), and so we

simply define $e(j)$ to be the largest value of i for which we've seen x_i in our scan.

Here's a final observation on this problem. Clearly, the solution looks very much like that of the Weighted Interval Scheduling Problem, and there's a fundamental reason for that. In fact, our billboard placement problem can be directly encoded as an instance of Weighted Interval Scheduling, as follows. Suppose that for each site x_i, we define an interval with endpoints $[x_i - 5, x_i]$ and weight r_i. Then, given any nonoverlapping set of intervals, the corresponding set of sites has the property that no two lie within 5 miles of each other. Conversely, given any such set of sites (no two within 5 miles), the intervals associated with them will be nonoverlapping. Thus the collections of nonoverlapping intervals correspond precisely to the set of valid billboard placements, and so dropping the set of intervals we've just defined (with their weights) into an algorithm for Weighted Interval Scheduling will yield the desired solution.

Solved Exercise 2

Through some friends of friends, you end up on a consulting visit to the cutting-edge biotech firm Clones 'R' Us (CRU). At first you're not sure how your algorithmic background will be of any help to them, but you soon find yourself called upon to help two identical-looking software engineers tackle a perplexing problem.

The problem they are currently working on is based on the *concatenation* of sequences of genetic material. If X and Y are each strings over a fixed alphabet $\mathcal{S}$, then XY denotes the string obtained by *concatenating* them—writing X followed by Y. CRU has identified a *target sequence* A of genetic material, consisting of m symbols, and they want to produce a sequence that is as similar to A as possible. For this purpose, they have a library $\mathcal{L}$ consisting of k (shorter) sequences, each of length at most n. They can cheaply produce any sequence consisting of copies of the strings in $\mathcal{L}$ concatenated together (with repetitions allowed).

Thus we say that a *concatenation* over $\mathcal{L}$ is any sequence of the form $B_1 B_2 \cdots B_\ell$, where each B_i belongs the set $\mathcal{L}$. (Again, repetitions are allowed, so B_i and B_j could be the same string in $\mathcal{L}$, for different values of i and j.) The problem is to find a concatenation over $\{B_i\}$ for which the sequence alignment cost is as small as possible. (For the purpose of computing the sequence alignment cost, you may assume that you are given a gap cost δ and a mismatch cost α_{pq} for each pair $p, q \in \mathcal{S}$.)

Give a polynomial-time algorithm for this problem.

Solution This problem is vaguely reminiscent of Segmented Least Squares: we have a long sequence of "data" (the string A) that we want to "fit" with shorter segments (the strings in $\mathcal{L}$).

If we wanted to pursue this analogy, we could search for a solution as follows. Let $B = B_1 B_2 \cdots B_\ell$ denote a concatenation over $\mathcal{L}$ that aligns as well as possible with the given string A. (That is, B is an optimal solution to the input instance.) Consider an optimal alignment M of A with B, let t be the first position in A that is matched with some symbol in B_ℓ, and let A_ℓ denote the substring of A from position t to the end. (See Figure 6.27 for an illustration of this with $\ell = 3$.) Now, the point is that in this optimal alignment M, the substring A_ℓ is optimally aligned with B_ℓ; indeed, if there were a way to better align A_ℓ with B_ℓ, we could substitute it for the portion of M that aligns A_ℓ with B_ℓ and obtain a better overall alignment of A with B.

This tells us that we can look at the optimal solution as follows. There's some final piece of A_ℓ that is aligned with one of the strings in $\mathcal{L}$, and for this piece all we're doing is finding the string in $\mathcal{L}$ that aligns with it as well as possible. Having found this optimal alignment for A_ℓ, we can break it off and continue to find the optimal solution for the remainder of A.

Thinking about the problem this way doesn't tell us exactly how to proceed—we don't know how long A_ℓ is supposed to be, or which string in $\mathcal{L}$ it should be aligned with. But this is the kind of thing we can search over in a dynamic programming algorithm. Essentially, we're in about the same spot we were in with the Segmented Least Squares Problem: there we knew that we had to break off some final subsequence of the input points, fit them as well as possible with one line, and then iterate on the remaining input points.

So let's set up things to make the search for A_ℓ possible. First, let $A[x : y]$ denote the substring of A consisting of its symbols from position x to position y, inclusive. Let $c(x, y)$ denote the cost of the optimal alignment of $A[x : y]$ with any string in $\mathcal{L}$. (That is, we search over each string in $\mathcal{L}$ and find the one that

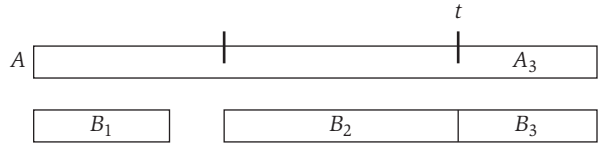


Figure 6.27 In the optimal concatentation of strings to align with A, there is a final string (B_3 in the figure) that aligns with a substring of A (A_3 in the figure) that extends from some position t to the end.

aligns best with $A[x : y]$.) Let $\text{OPT}(j)$ denote the alignment cost of the optimal solution on the string $A[1 : j]$.

The argument above says that an optimal solution on $A[1 : j]$ consists of identifying a final "segment boundary" $t < j$, finding the optimal alignment of $A[t : j]$ with a single string in $\mathcal{L}$, and iterating on $A[1 : t - 1]$. The cost of this alignment of $A[t : j]$ is just $c(t, j)$, and the cost of aligning with what's left is just $\text{OPT}(t - 1)$. This suggests that our subproblems fit together very nicely, and it justifies the following recurrence.

(6.37) $\text{OPT}(j) = \min_{t<j} c(t, j) + \text{OPT}(t - 1)$ *for* $j \geq 1$, *and* $\text{OPT}(0) = 0$.

The full algorithm consists of first computing the quantities $c(t, j)$, for $t < j$, and then building up the values $\text{OPT}(j)$ in order of increasing j. We hold these values in an array M.

```
Set M[0] = 0
For all pairs 1 ≤ t ≤ j ≤ m
  Compute the cost c(t, j) as follows:
  For each string B ∈ ℒ
    Compute the optimal alignment of B with A[t : j]
  Endfor
  Choose the B that achieves the best alignment, and use
    this alignment cost as c(t, j)
Endfor
For j = 1, 2, . . . , n
  Use the recurrence (6.37) to compute M[j]
Endfor
Return M[n]
```

As usual, we can get a concatentation that achieves it by tracing back over the array of OPT values.

Let's consider the running time of this algorithm. First, there are $O(m^2)$ values $c(t, j)$ that need to be computed. For each, we try each string of the k strings $B \in \mathcal{L}$, and compute the optimal alignment of B with $A[t : j]$ in time $O(n(j - t)) = O(mn)$. Thus the total time to compute all $c(t, j)$ values is $O(km^3n)$.

This dominates the time to compute all OPT values: Computing $\text{OPT}(j)$ uses the recurrence in (6.37), and this takes $O(m)$ time to compute the minimum. Summing this over all choices of $j = 1, 2, \ldots, m$, we get $O(m^2)$ time for this portion of the algorithm.